
Data Mining and Machine Learning (CSE 321)

Topic – 8: Cluster Analysis

<https://chatgpt.com/c/69f65e5c-1ecc-83a2-b07b-eea9cb1df765>

Topic Contents



Overview



K-means



Hierarchical Clustering

Recommended Reading



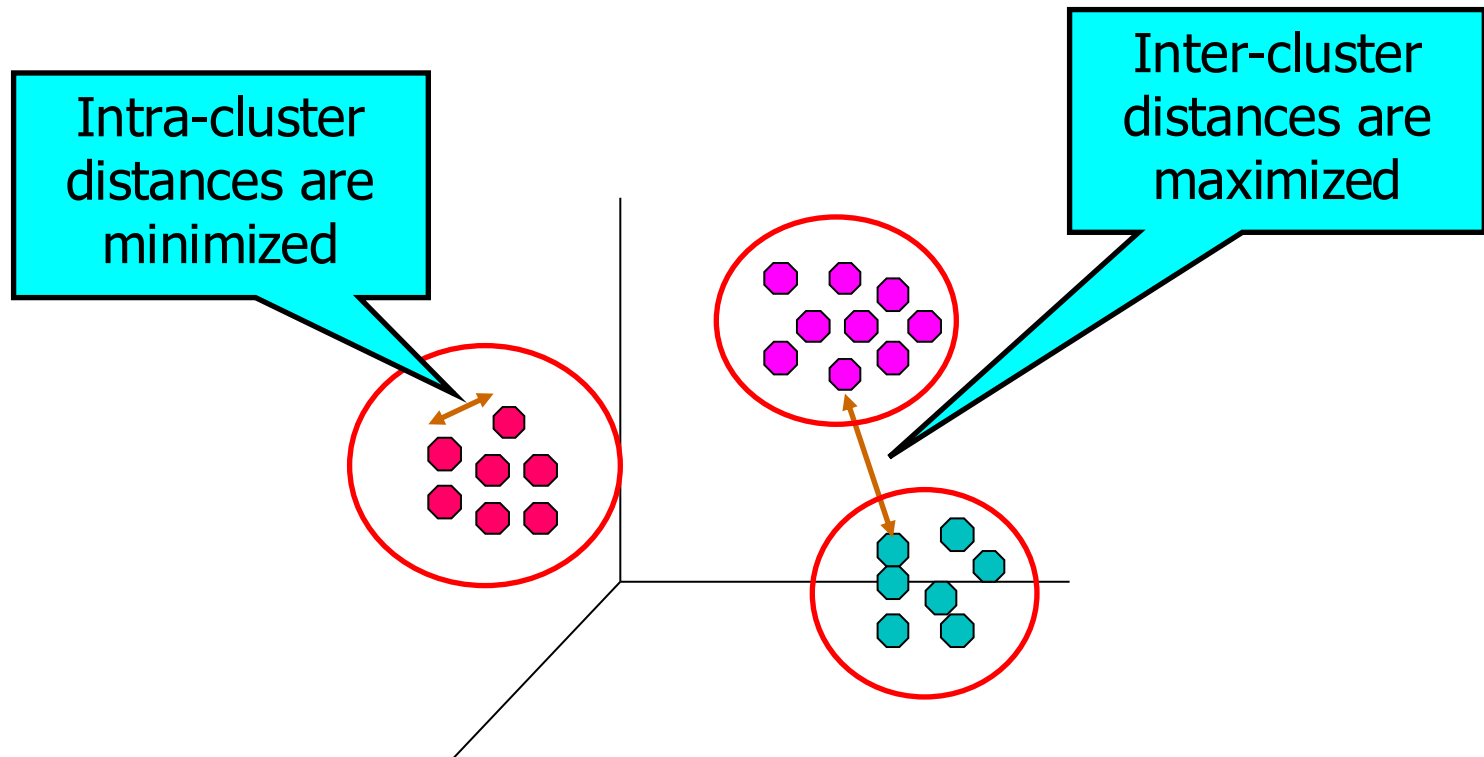
"Introduction to Data Mining," Pang-Ning Tan, Michael Steinbach and Vipin Kumar, Addison Wesley, 2006.



Chapter 8 (Cluster Analysis: Basic Concepts and Algorithms)

What is Cluster Analysis?

- Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups



Applications of Cluster Analysis

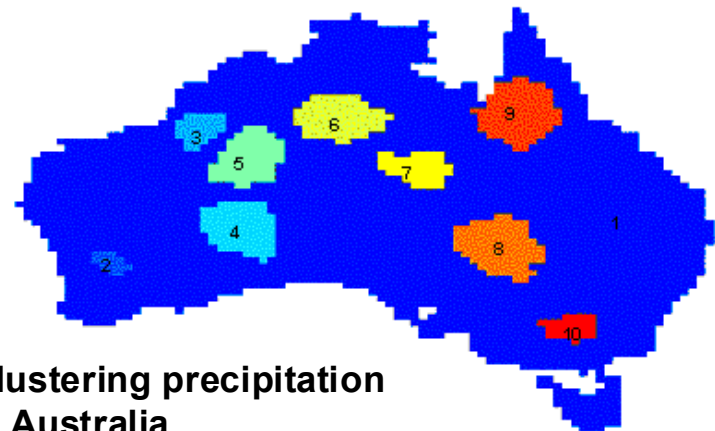
□ Understanding

- Group related documents for browsing, group genes and proteins that have similar functionality, or group stocks with similar price fluctuations

	<i>Discovered Clusters</i>	<i>Industry Group</i>
1	Applied-Matl-DOWN,Bay-Network-DOWN,3-COM-DOWN,Cabletron-Sys-DOWN,CISCO-DOWN,HP-DOWN,DSC-Comm-DOWN,INTEL-DOWN,LSI-Logic-DOWN,Micron-Tech-DOWN,Texas-Inst-DOWN,Tellabs-Inc-DOWN,Natl-Semiconduct-DOWN,Oracl-DOWN,SGI-DOWN,Sun-DOWN	Technology1-DOWN
2	Apple-Comp-DOWN,Autodesk-DOWN,DEC-DOWN,ADV-Micro-Device-DOWN,Andrew-Corp-DOWN,Computer-Assoc-DOWN,Circuit-City-DOWN,Compaq-DOWN,EMC-Corp-DOWN,Gen-Inst-DOWN,Motorola-DOWN,Microsoft-DOWN,Scientific-Atl-DOWN	Technology2-DOWN
3	Fannie-Mae-DOWN,Fed-Home-Loan-DOWN,MBNA-Corp-DOWN,Morgan-Stanley-DOWN	Financial-DOWN
4	Baker-Hughes-UP,Dresser-Inds-UP,Halliburton-HLD-UP,Louisiana-Land-UP,Phillips-Petro-UP,Unocal-UP,Schlumberger-UP	Oil-UP

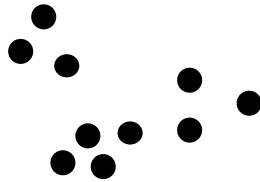
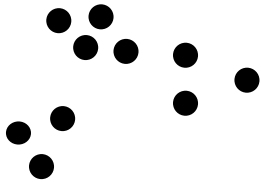
□ Summarization

- Reduce the size of large data sets

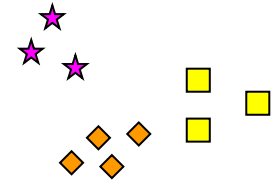
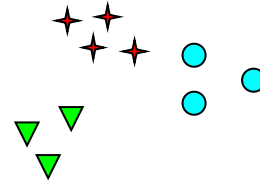


Clustering precipitation
in Australia

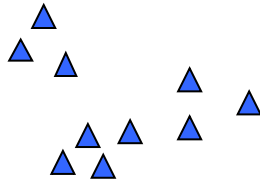
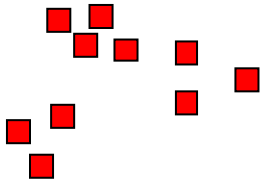
Notion of a Cluster can be Ambiguous



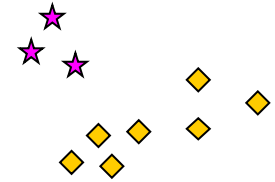
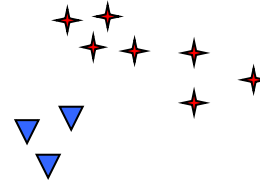
How many clusters?



Six Clusters



Two Clusters

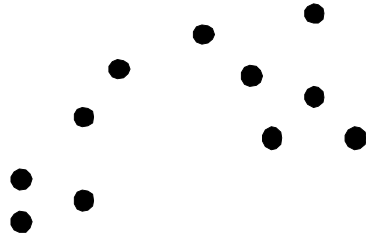


Four Clusters

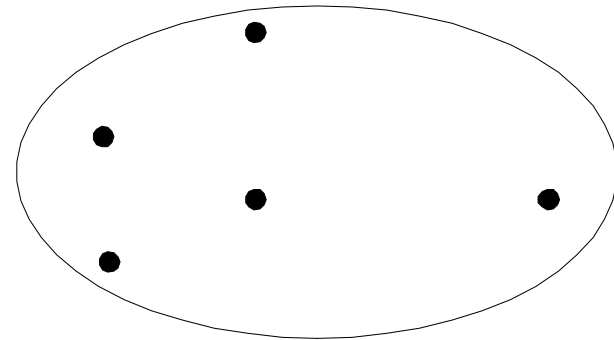
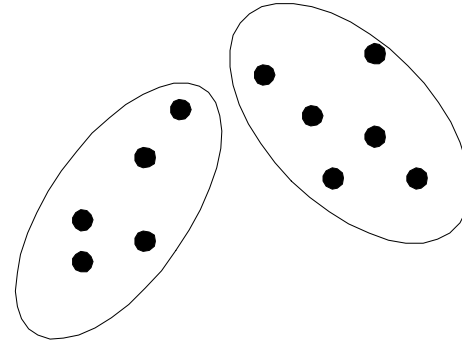
Types of Clusterings

- A **clustering** is a set of clusters
- Important distinction between **hierarchical** and **partitional** sets of clusters
- Partitional Clustering
 - A division data objects into non-overlapping subsets (clusters) such that each data object is in exactly one subset
- Hierarchical clustering
 - A set of nested clusters organized as a hierarchical tree

Partitional Clustering

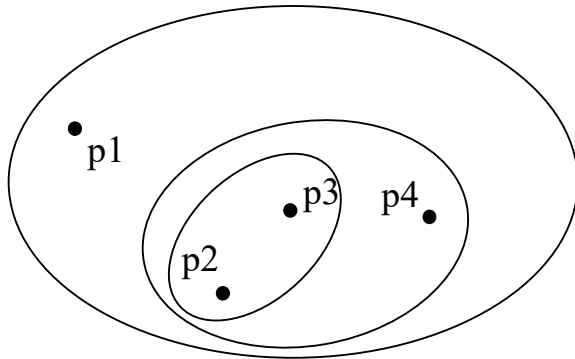


Original Points

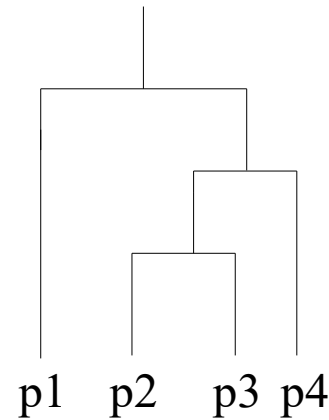


A Partitional Clustering

Hierarchical Clustering



Traditional Hierarchical Clustering



Traditional Dendrogram

Other Distinctions Between Sets of Clusters

□ Exclusive versus non-exclusive

- In non-exclusive clusterings, points may belong to multiple clusters.
- Can represent multiple classes or ‘border’ points

□ Fuzzy versus non-fuzzy

- In fuzzy clustering, a point belongs to every cluster with some weight between 0 and 1
- Weights must sum to 1
- Probabilistic clustering has similar characteristics

□ Partial versus complete

- In some cases, we only want to cluster some of the data

□ Heterogeneous versus homogeneous

- Cluster of widely different sizes, shapes, and densities

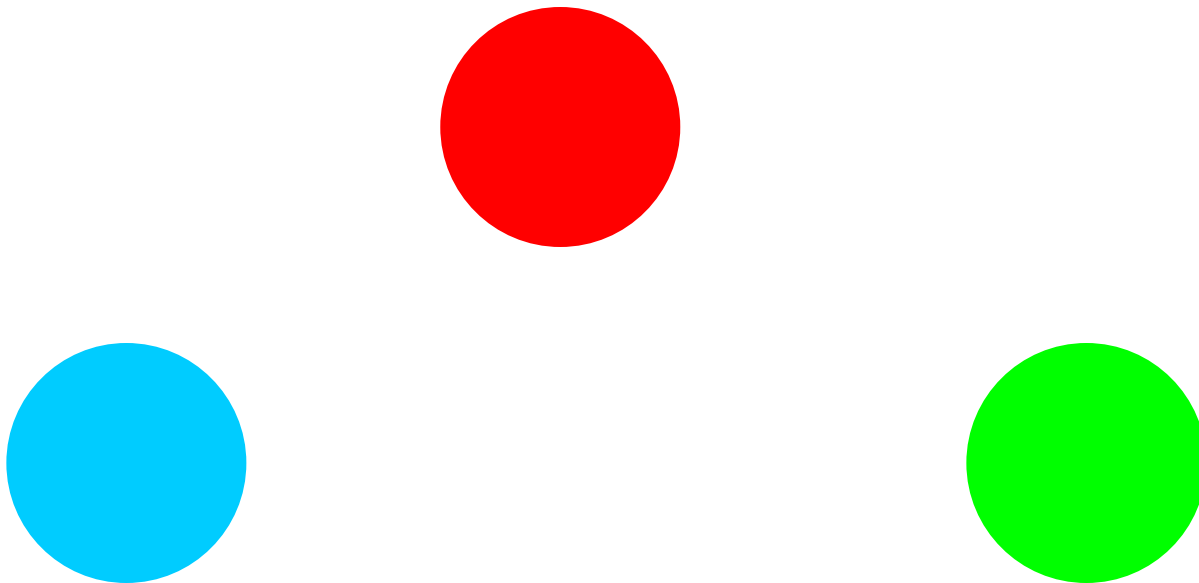
Types of Clusters

- Well-separated clusters
- Center-based clusters
- Contiguous clusters
- Density-based clusters
- Property or Conceptual
- Described by an Objective Function

Types of Clusters: Well-Separated

□ Well-Separated Clusters:

- A cluster is a set of points such that any point in a cluster is closer (or more similar) to every other point in the cluster than to any point not in the cluster.

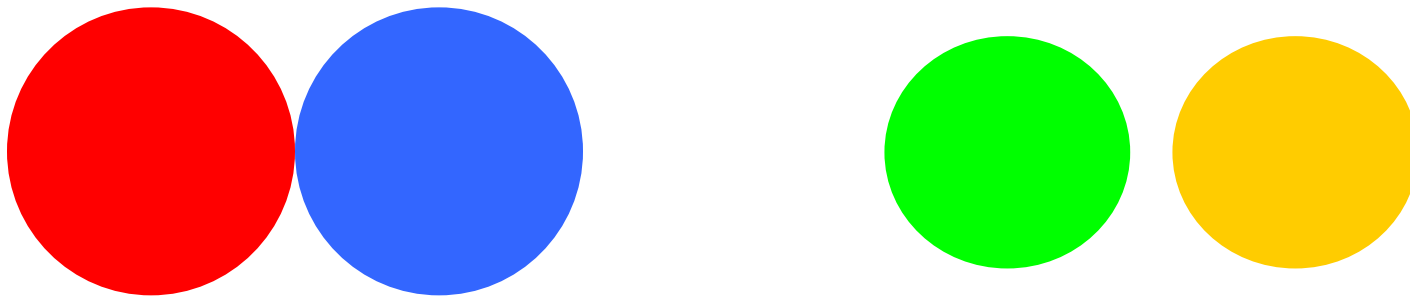


3 well-separated clusters

Types of Clusters: Center-Based

□ Prototype/Center-based

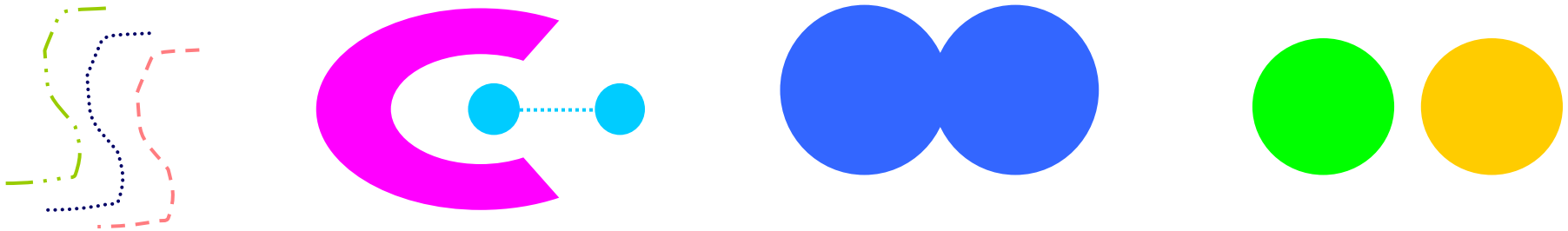
- A cluster is a set of objects such that an object in a cluster is closer (more similar) to the (prototype) (e.g., “center”) of a cluster, than to the center of any other cluster
- The center of a cluster is often a **centroid**, the average of all the points in the cluster, or a **medoid**, the most “representative” point of a cluster



4 center-based clusters

Types of Clusters: Contiguity-Based

- Contiguous Cluster (Nearest neighbor or Transitive)
 - A cluster is a set of points such that a point in a cluster is closer (or more similar) to at least one point in its cluster than to any point not in the cluster.

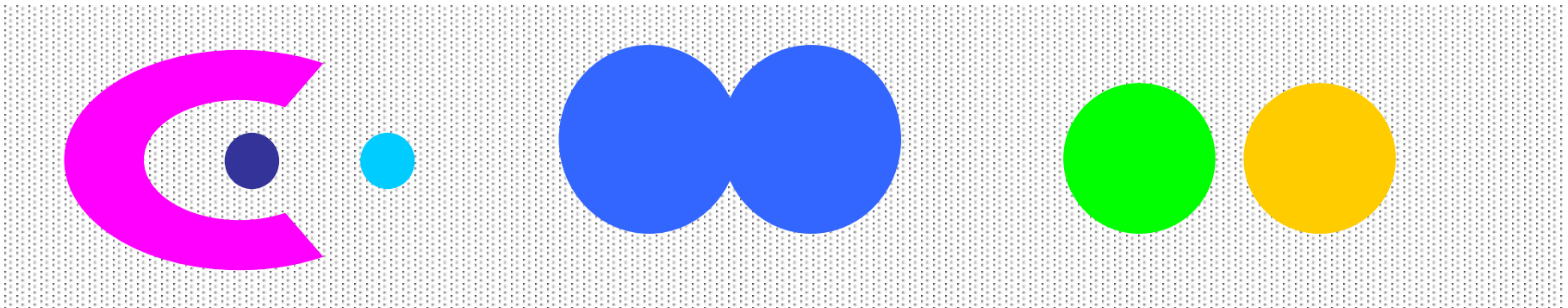


8 contiguous clusters

Types of Clusters: Density-Based

□ Density-based

- A cluster is a dense region of points, which is separated by low-density regions, from other regions of high density.
- Used when the clusters are irregular or intertwined, and when noise and outliers are present.



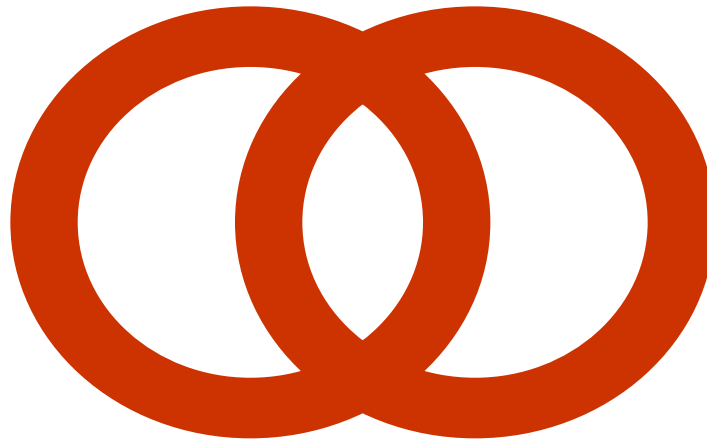
6 density-based clusters

Types of Clusters: Conceptual Clusters

□ Shared Property or Conceptual Clusters

- Finds clusters that share some common property or represent a particular concept.

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2 Overlapping Circles

Clustering Algorithms

- K-means and its variants
- Hierarchical clustering
- Density-based clustering

K-means Clustering

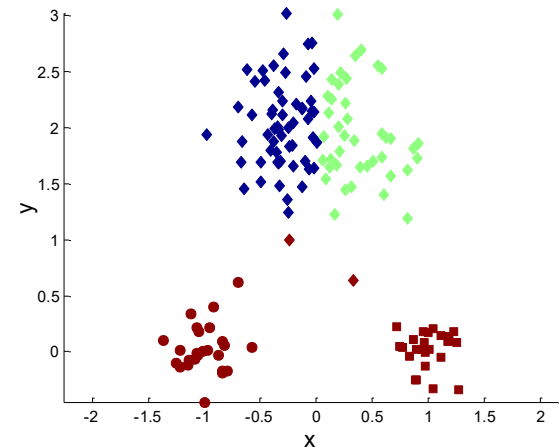
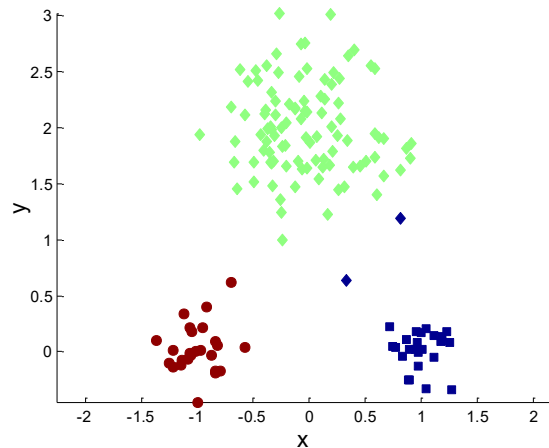
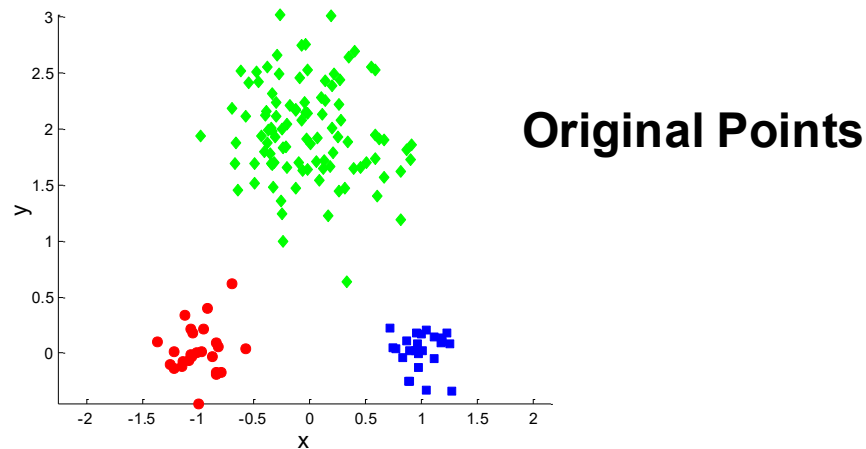
- Partitional clustering approach
- Each cluster is associated with a **centroid** (center point)
- Each point is assigned to the cluster with the closest centroid
- Number of clusters, K , must be specified
- The basic algorithm is very simple

-
- 1: Select K points as the initial centroids.
 - 2: **repeat**
 - 3: Form K clusters by assigning all points to the closest centroid.
 - 4: Recompute the centroid of each cluster.
 - 5: **until** The centroids don't change
-

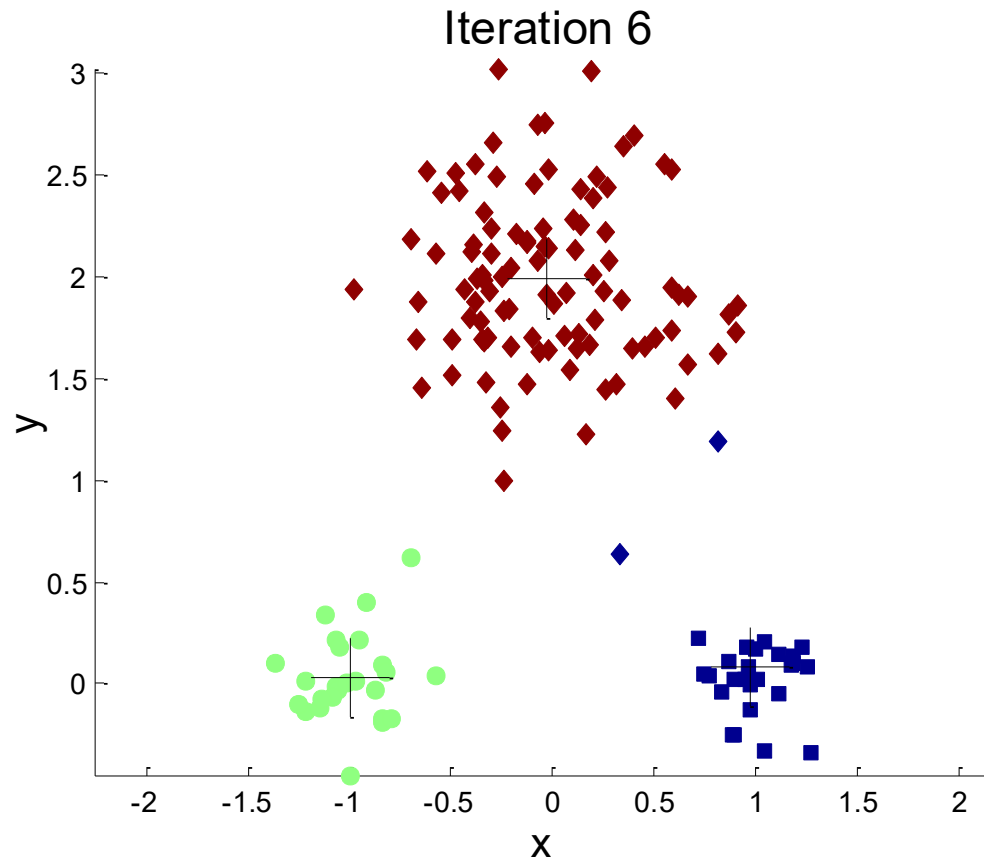
K-means Clustering – Details

- Initial centroids are often chosen randomly.
 - Clusters produced vary from one run to another.
- The centroid is (typically) the mean of the points in the cluster.
- ‘Closeness’ is measured by Euclidean distance, cosine similarity, correlation, etc.
- K-means will converge for common similarity measures mentioned above.
- Most of the convergence happens in the first few iterations.
 - Often the stopping condition is changed to ‘Until relatively few points change clusters’
- Complexity is $O(n * K * I * d)$
 - n = number of points, K = number of clusters,
 I = number of iterations, d = number of attributes

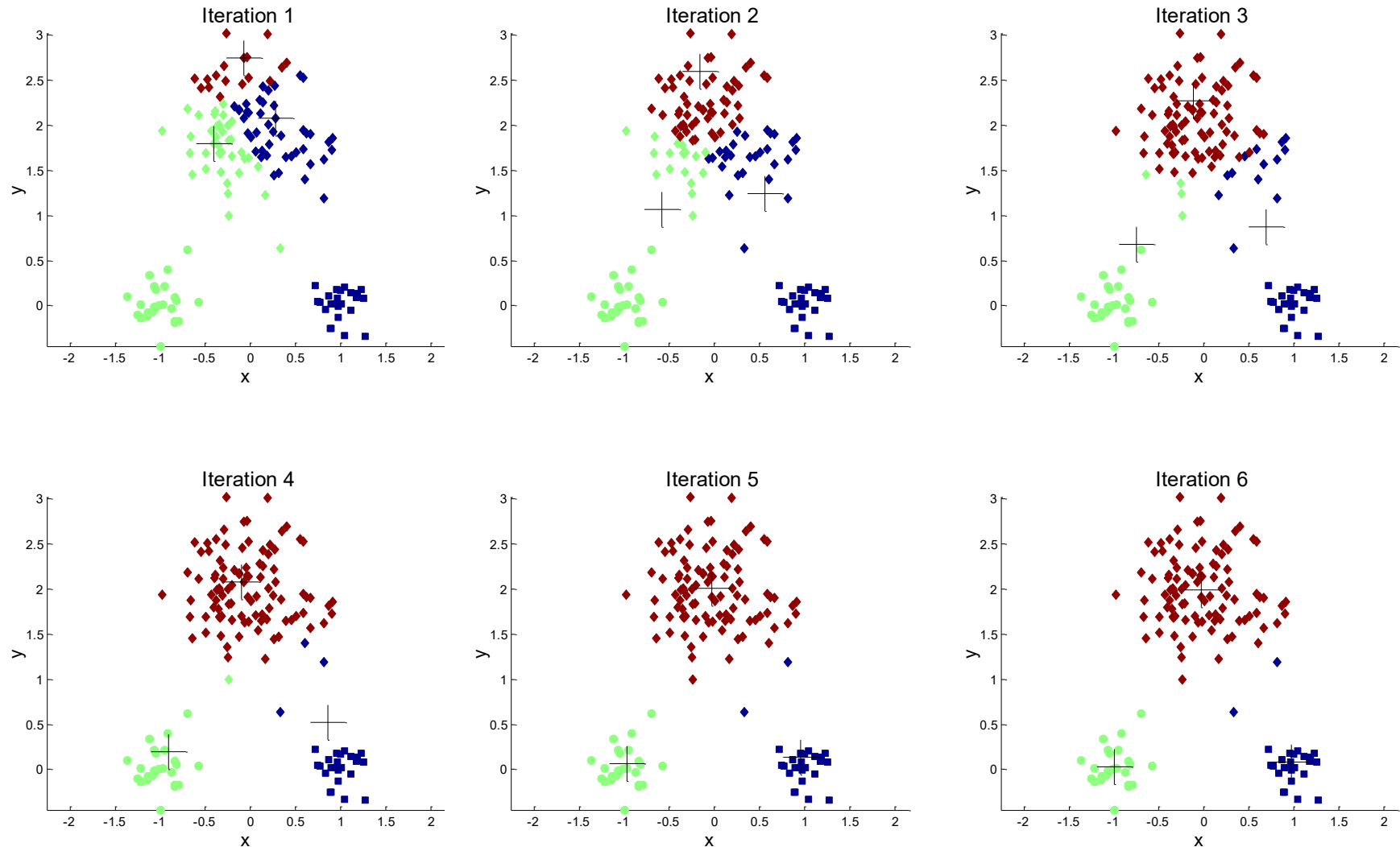
Two different K-means Clusterings



Importance of Choosing Initial Centroids



Importance of Choosing Initial Centroids



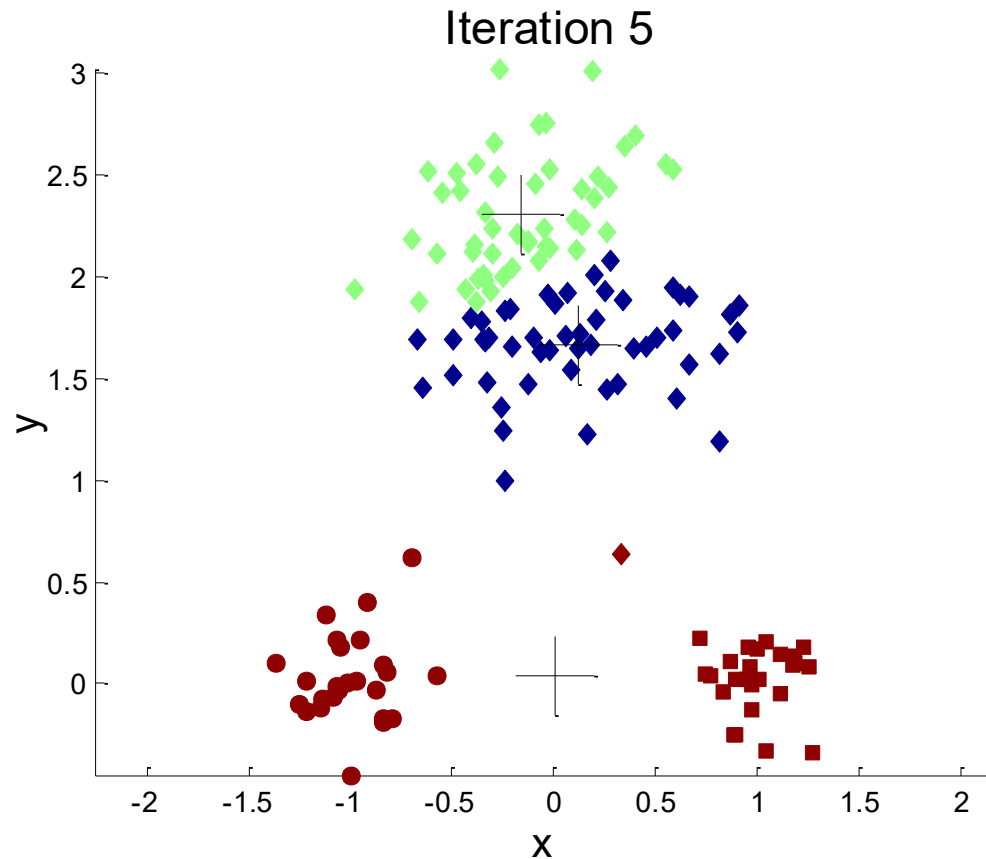
Evaluating K-means Clusters

- Most common measure is Sum of Squared Error (SSE)
 - For each point, the error is the distance to the nearest cluster
 - To get SSE, we square these errors and sum them.

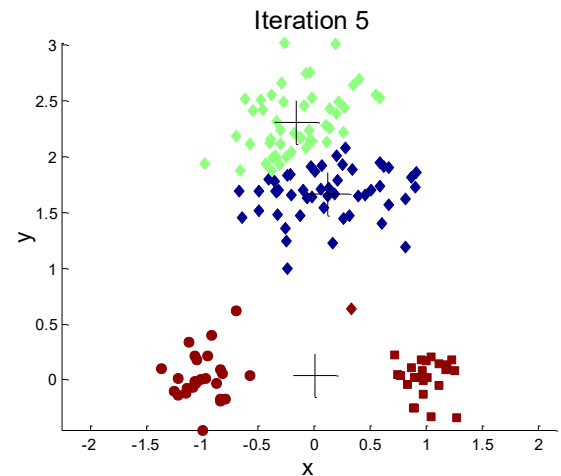
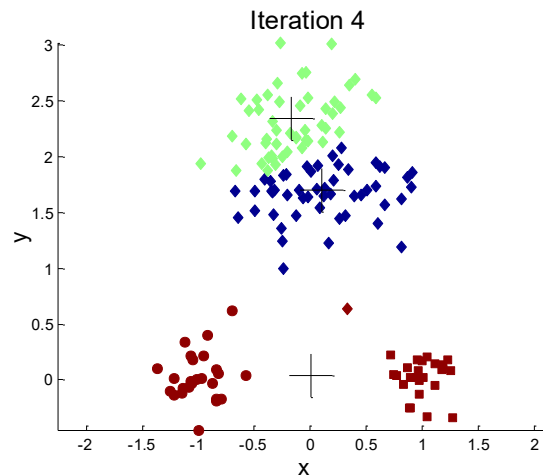
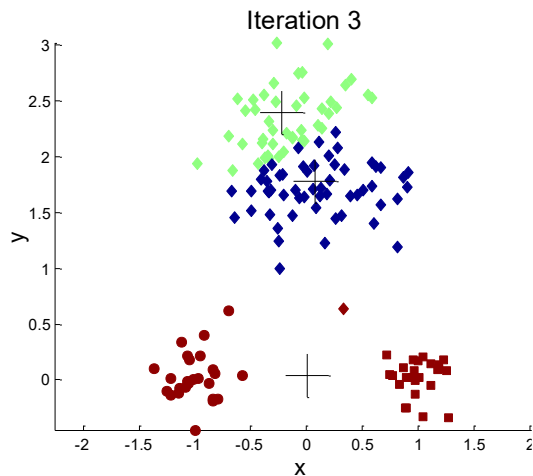
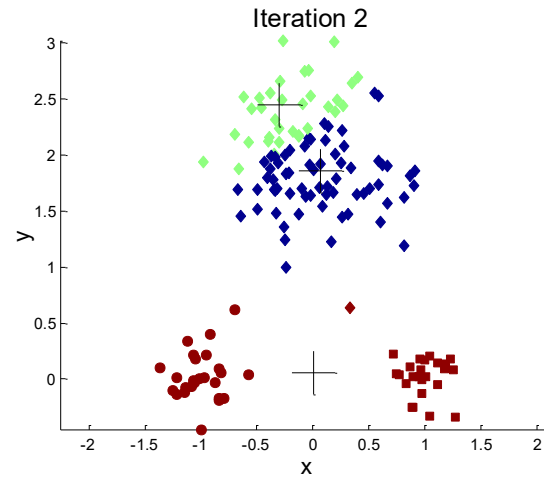
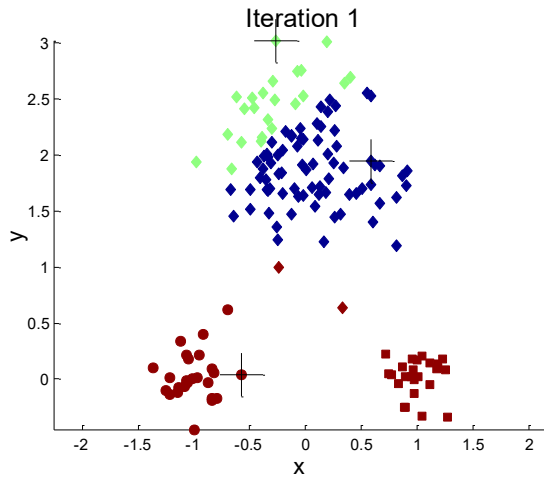
$$SSE = \sum_{i=1}^K \sum_{x \in C_i} dist^2(m_i, x)$$

- x is a data point in cluster C_i and m_i is the representative point for cluster C_i
 - ◆ can show that m_i corresponds to the center (mean) of the cluster
- Given two clusters, we can choose the one with the smallest error
- One easy way to reduce SSE is to increase K , the number of clusters
 - ◆ A good clustering with smaller K can have a lower SSE than a poor clustering with higher K

Importance of Choosing Initial Centroids ...



Importance of Choosing Initial Centroids ...



Solutions to Initial Centroids Problem

- Multiple runs
 - Helps, but probability is not on your side
- Sample and use hierarchical clustering to determine initial centroids
- Select more than k initial centroids and then select among these initial centroids
 - Select most widely separated
- Postprocessing
- Bisecting K-means
 - Not as susceptible to initialization issues

Handling Empty Clusters

- Basic K-means algorithm can yield empty clusters
- Several strategies
 - Choose the point that contributes most to SSE
 - Choose a point from the cluster with the highest SSE
 - If there are several empty clusters, the above can be repeated several times.

Updating Centers Incrementally

- In the basic K-means algorithm, centroids are updated after all points are assigned to a centroid
- An alternative is to update the centroids after each assignment (incremental approach)
 - Each assignment updates zero or two centroids
 - More expensive
 - Introduces an order dependency
 - Never get an empty cluster
 - Can use “weights” to change the impact

Pre-processing and Post-processing

□ Pre-processing

- Normalize the data
- Eliminate outliers

□ Post-processing

- Eliminate small clusters that may represent outliers
- Split 'loose' clusters, i.e., clusters with relatively high SSE
- Merge clusters that are 'close' and that have relatively low SSE

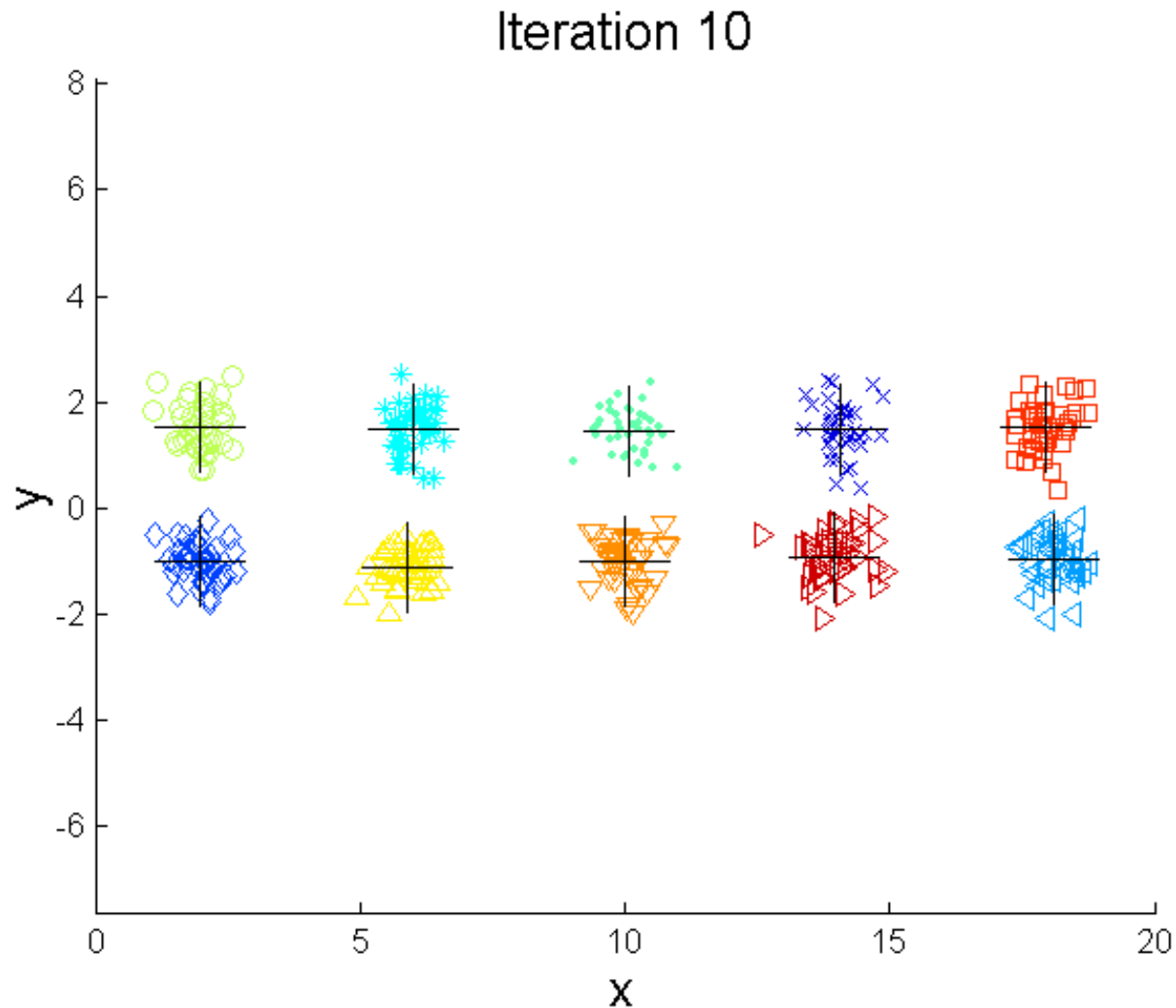
Bisecting K-means

□ Bisecting K-means algorithm

- Variant of K-means

```
1: Initialize the list of clusters to contain the cluster containing all points.  
2: repeat  
3:   Select a cluster from the list of clusters  
4:   for  $i = 1$  to number_of_iterations do  
5:     Bisect the selected cluster using basic K-means  
6:   end for  
7:   Add the two clusters from the bisection with the lowest SSE to the list of clusters.  
8: until Until the list of clusters contains  $K$  clusters
```

Bisecting K-means Example

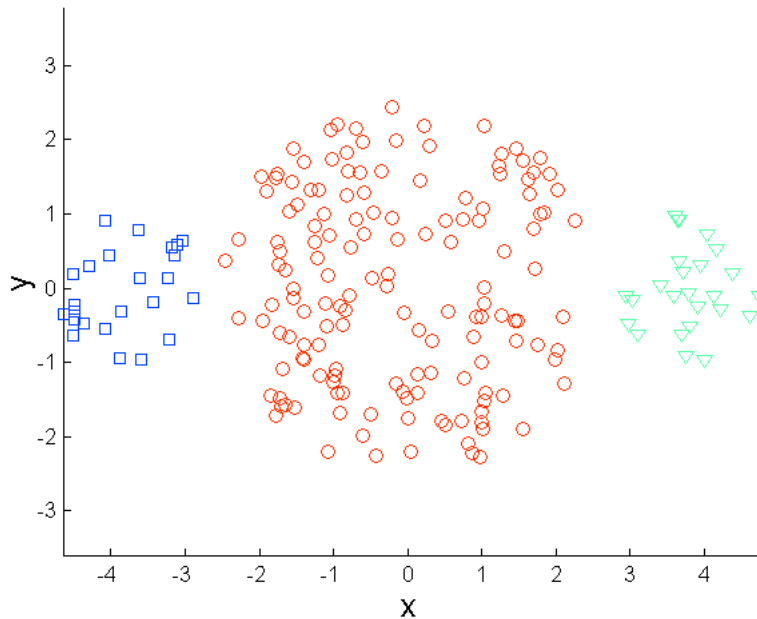


Limitations of K-means

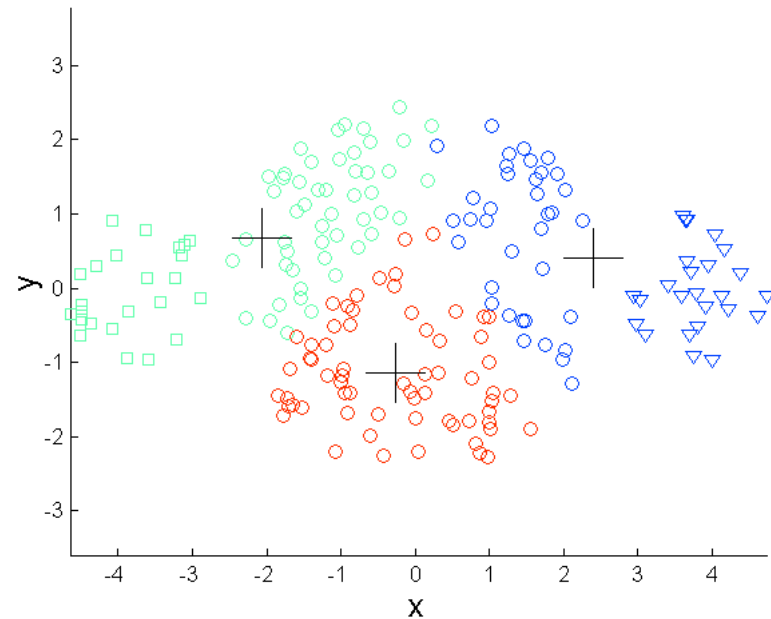
- K-means has problems when clusters are of differing
 - Sizes
 - Densities
 - Non-globular shapes

- K-means has problems when the data contains outliers.

Limitations of K-means: Differing Sizes

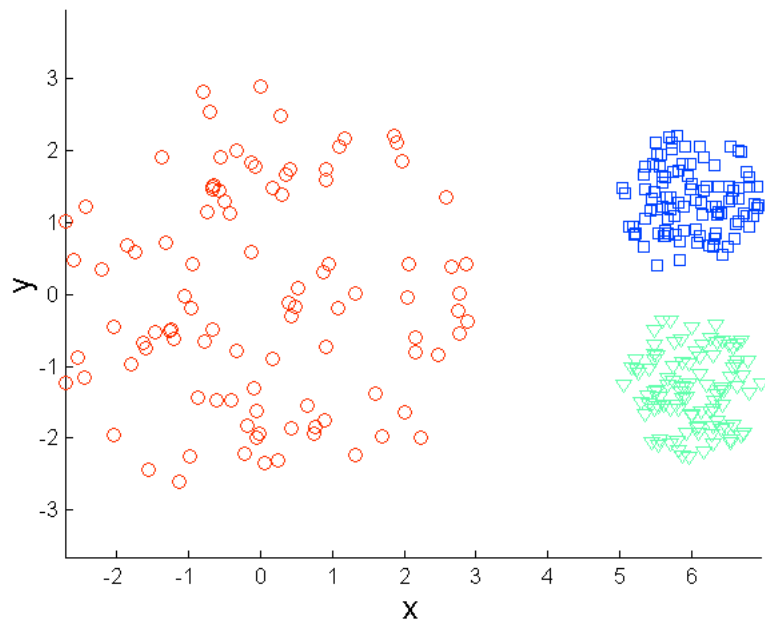


Original Points

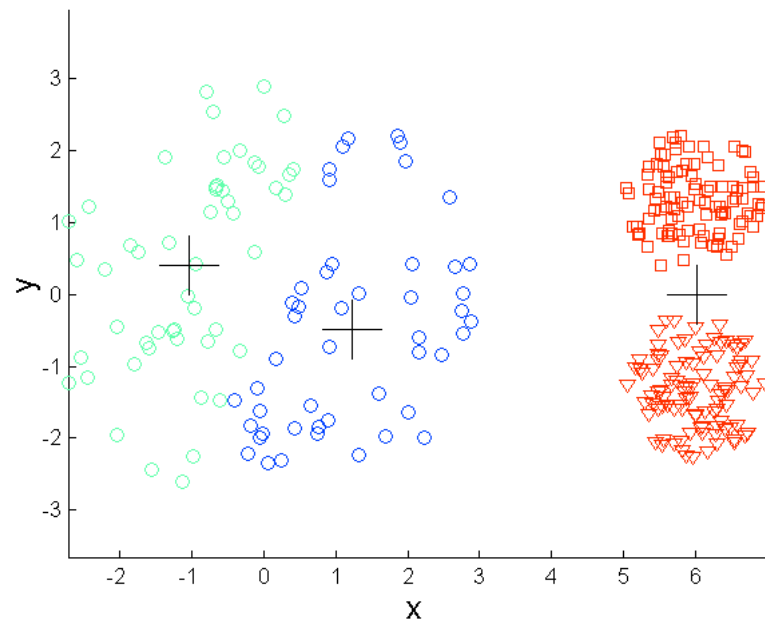


K-means (3 Clusters)

Limitations of K-means: Differing Density

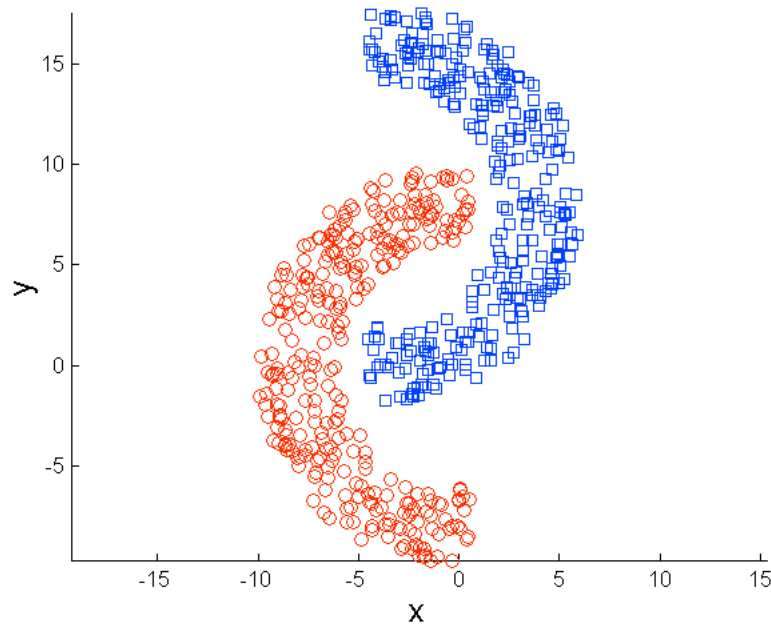


Original Points

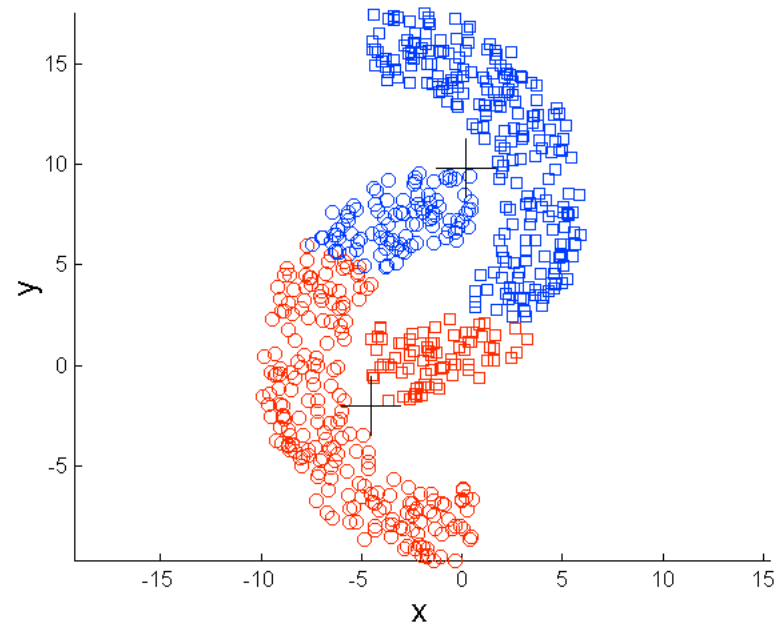


K-means (3 Clusters)

Limitations of K-means: Non-globular Shapes

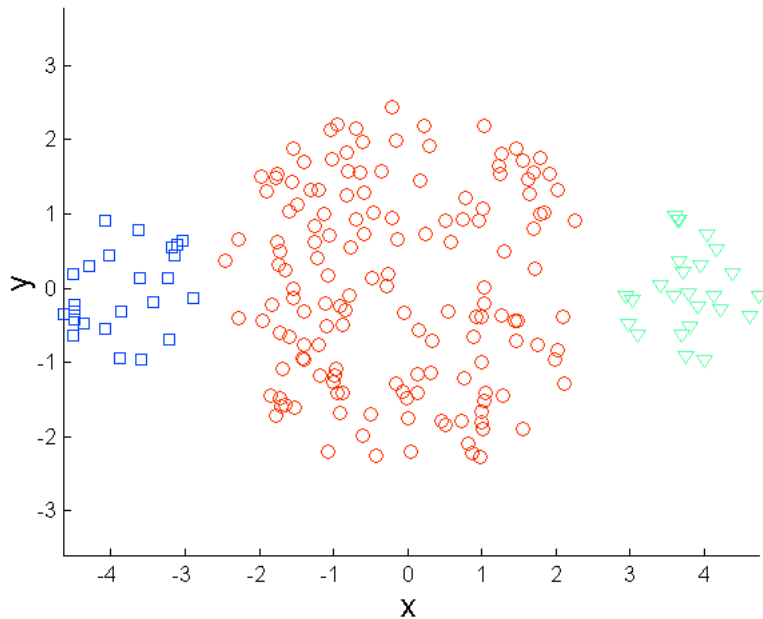


Original Points

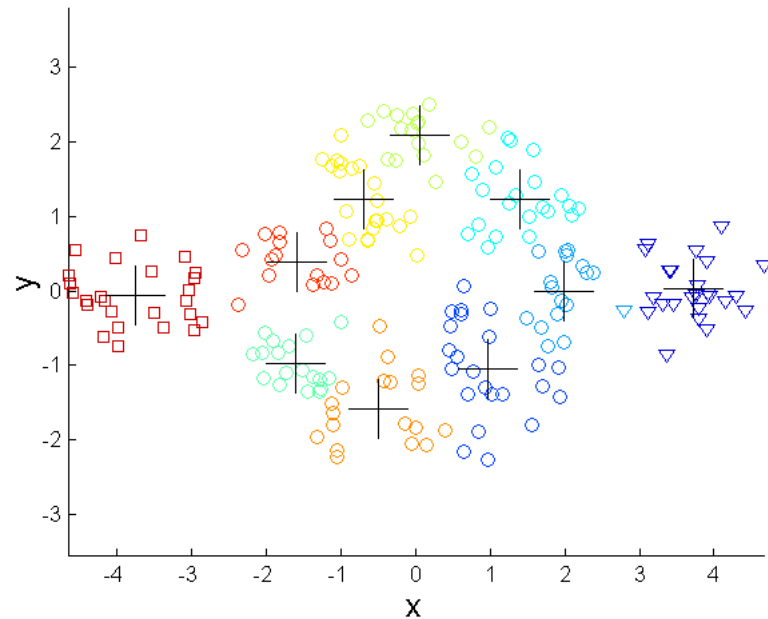


K-means (2 Clusters)

Overcoming K-means Limitations



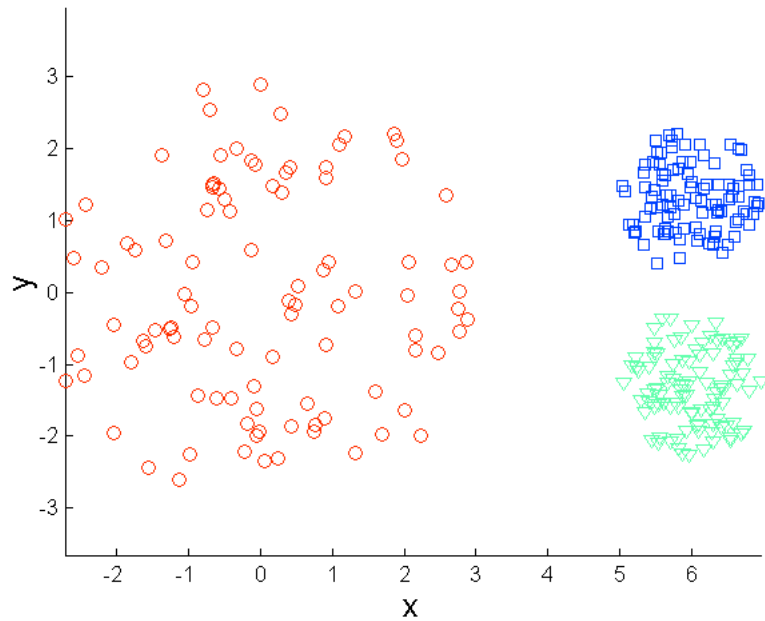
Original Points



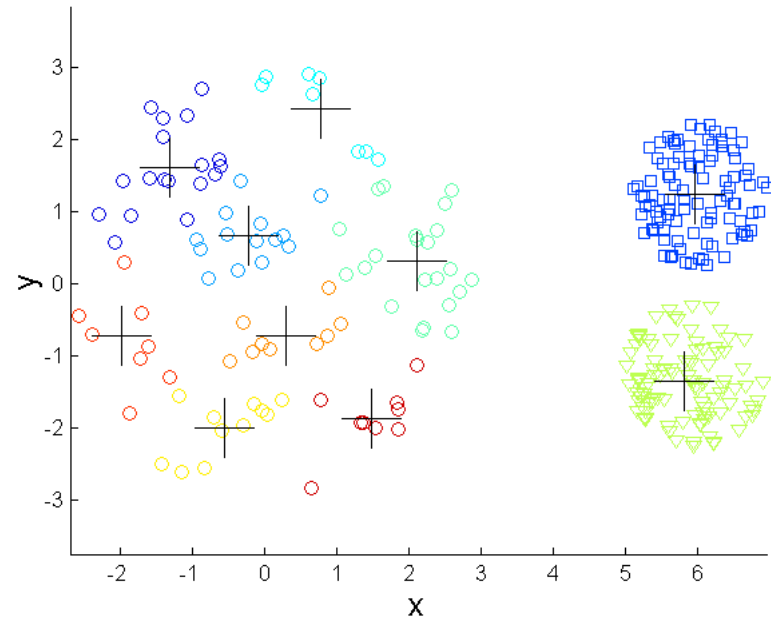
K-means Clusters

One solution is to use many clusters.
Find parts of clusters, but need to put together.

Overcoming K-means Limitations

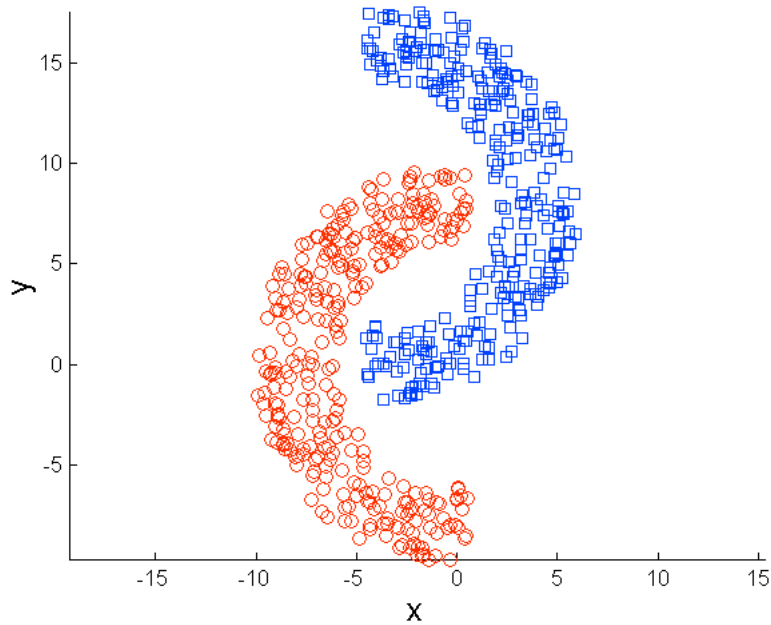


Original Points

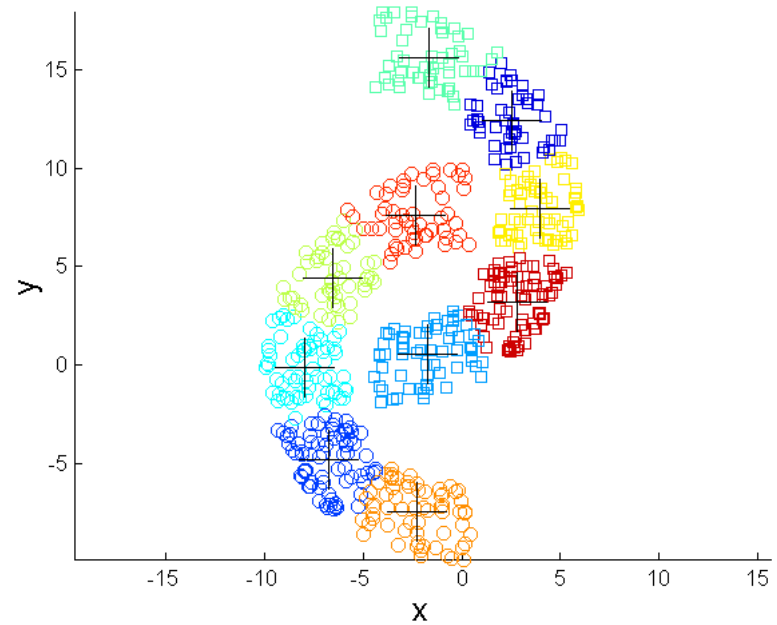


K-means Clusters

Overcoming K-means Limitations



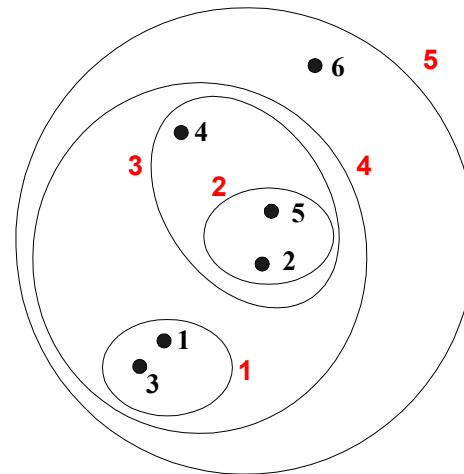
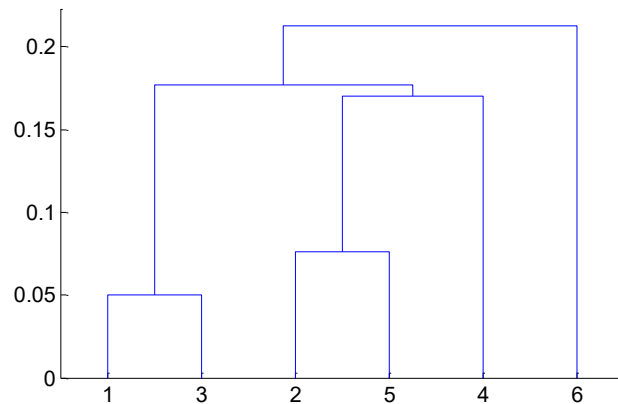
Original Points



K-means Clusters

Hierarchical Clustering

- Produces a set of nested clusters organized as a hierarchical tree
- Can be visualized as a dendrogram
 - A tree like diagram that records the sequences of merges or splits



Strengths of Hierarchical Clustering

- Do not have to assume any particular number of clusters
 - Any desired number of clusters can be obtained by 'cutting' the dendrogram at the proper level
- They may correspond to meaningful taxonomies
 - Example in biological sciences (e.g., animal kingdom, phylogeny reconstruction, ...)

Hierarchical Clustering

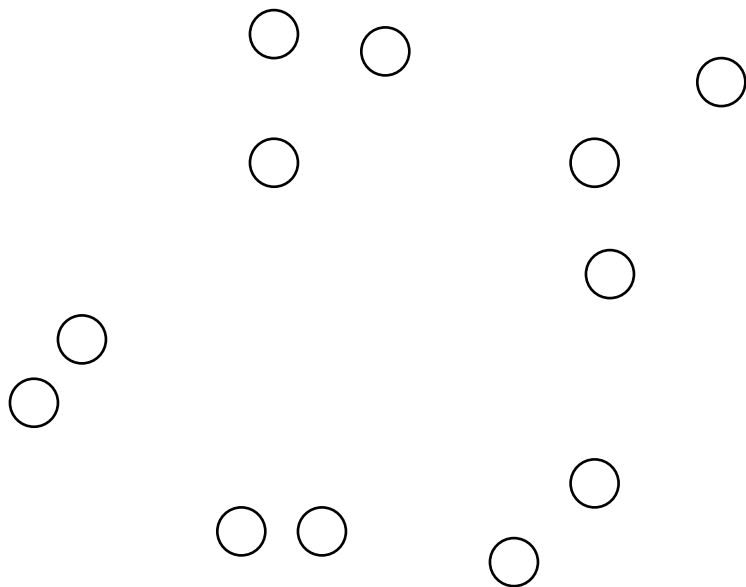
- Two main types of hierarchical clustering
 - Agglomerative:
 - ◆ Start with the points as individual clusters
 - ◆ At each step, merge the closest pair of clusters until only one cluster (or k clusters) left
 - Divisive:
 - ◆ Start with one, all-inclusive cluster
 - ◆ At each step, split a cluster until each cluster contains a point (or there are k clusters)
- Traditional hierarchical algorithms use a similarity or distance matrix
 - Merge or split one cluster at a time

Agglomerative Clustering Algorithm

- More popular hierarchical clustering technique
- Basic algorithm is straightforward
 1. Compute the proximity matrix
 2. Let each data point be a cluster
 3. **Repeat**
 4. Merge the two closest clusters
 5. Update the proximity matrix
 6. **Until** only a single cluster remains
- Key operation is the computation of the proximity of two clusters
 - Different approaches to defining the distance between clusters distinguish the different algorithms

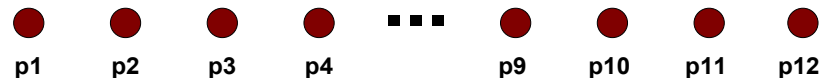
Starting Situation

- Start with clusters of individual points and a proximity matrix



	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

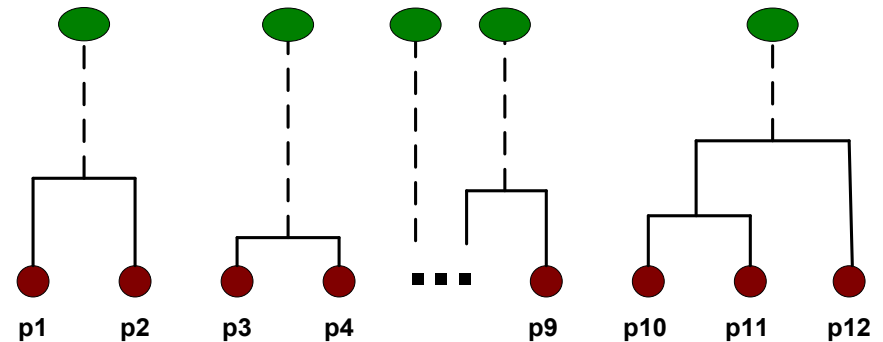
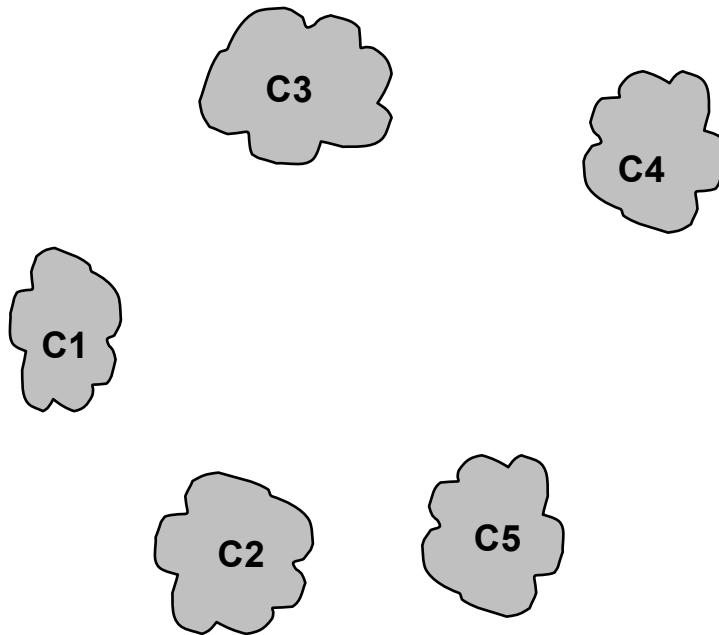


Intermediate Situation

- After some merging steps, we have some clusters

	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

Proximity Matrix

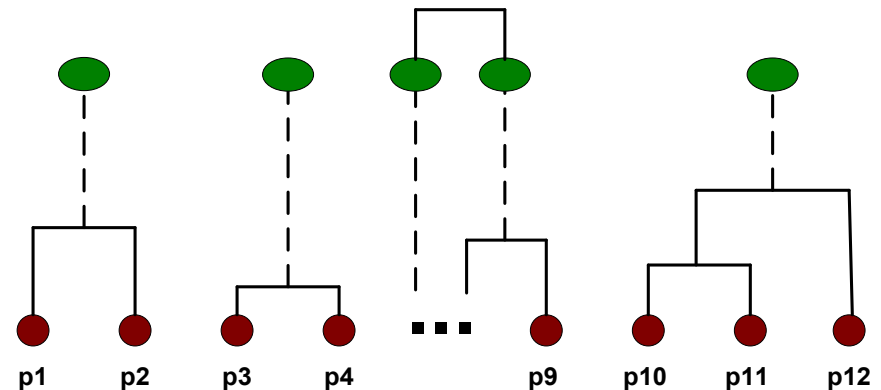
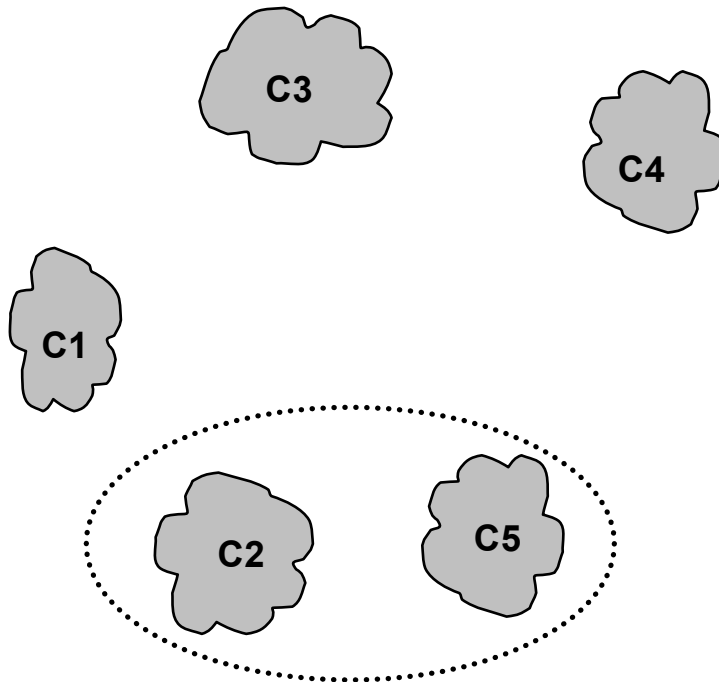


Intermediate Situation

- We want to merge the two closest clusters (C2 and C5) and update the proximity matrix.

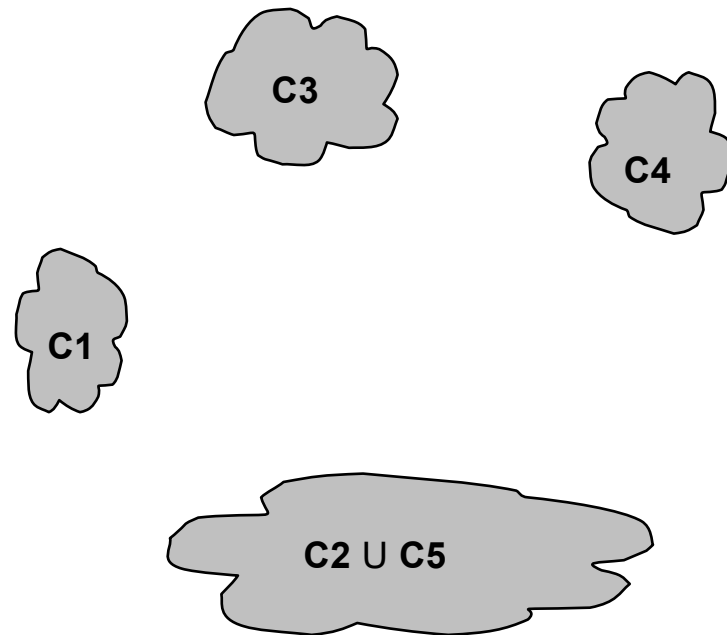
	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

Proximity Matrix



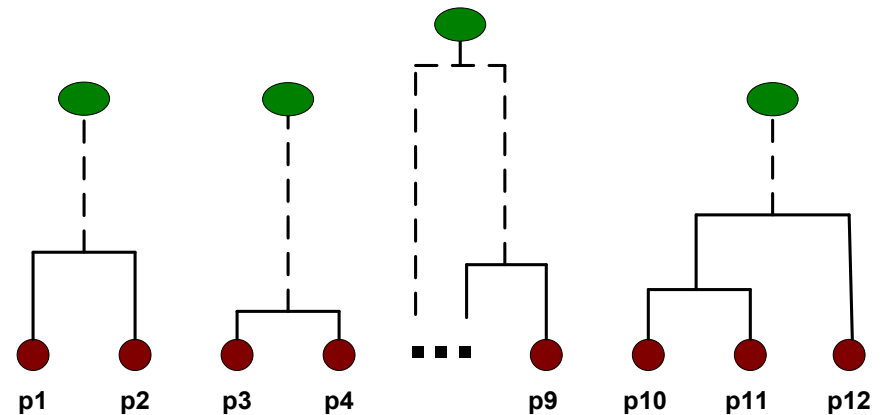
After Merging

- The question is “How do we update the proximity matrix?”

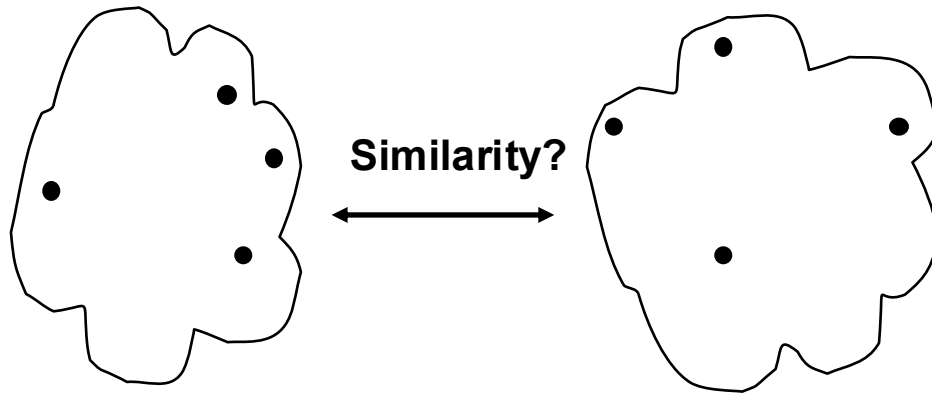


		C2 U C5			
		C1	C5	C3	C4
C1			?		
C2 U C5		?	?	?	?
C3			?		
C4			?		

Proximity Matrix



How to Define Inter-Cluster Similarity

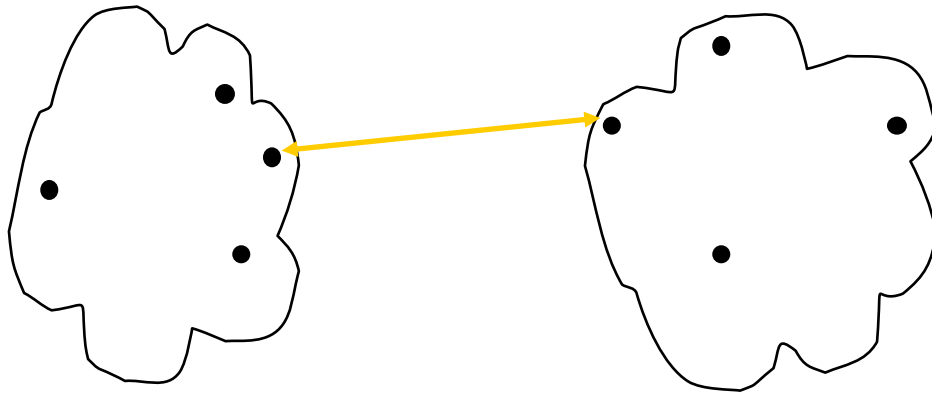


- MIN
- MAX
- Group Average
- Distance Between Centroids
- Other methods driven by an objective function
 - Ward's Method uses squared error

	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

How to Define Inter-Cluster Similarity

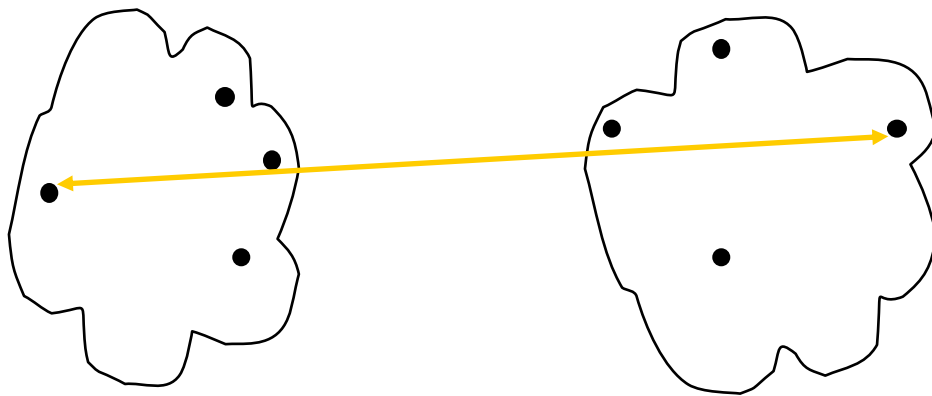


- MIN
- MAX
- Group Average
- Distance Between Centroids
- Other methods driven by an objective function
 - Ward's Method uses squared error

	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

How to Define Inter-Cluster Similarity

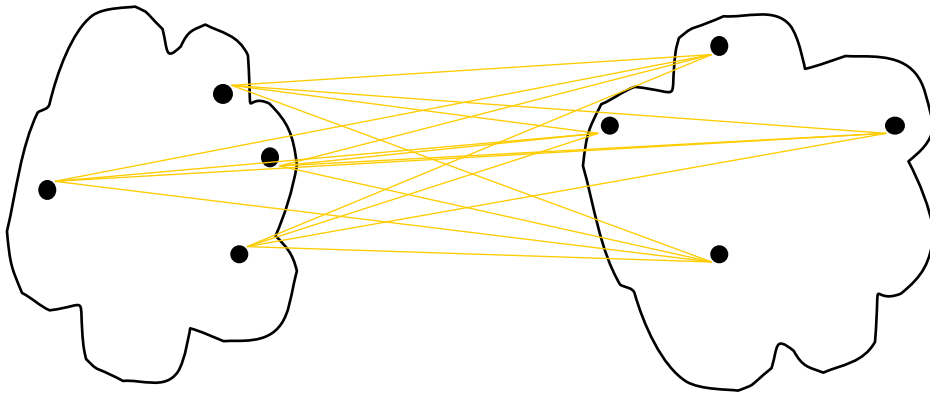


- MIN
- MAX
- Group Average
- Distance Between Centroids
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 - Ward's Method uses squared error

	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

How to Define Inter-Cluster Similarity

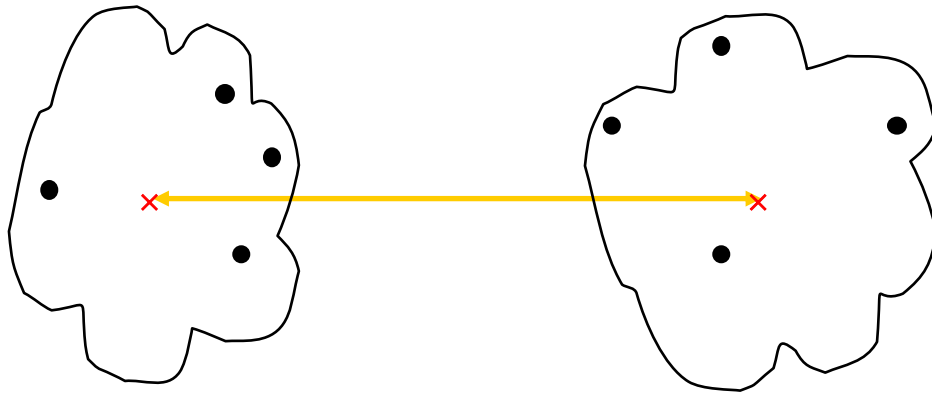


- MIN
- MAX
- **Group Average**
- Distance Between Centroids
- Other methods driven by an objective function
 - Ward's Method uses squared error

	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

How to Define Inter-Cluster Similarity



- MIN
- MAX
- Group Average
- Distance Between Centroids
- Other methods driven by an objective function
 - Ward's Method uses squared error

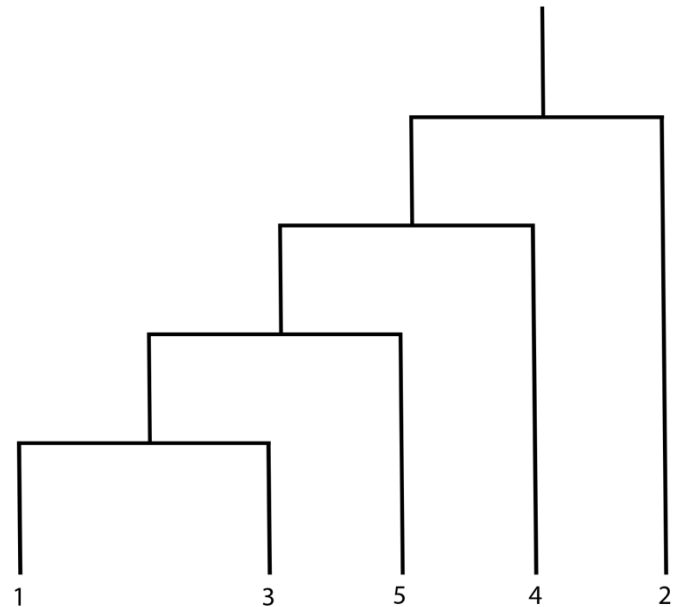
	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

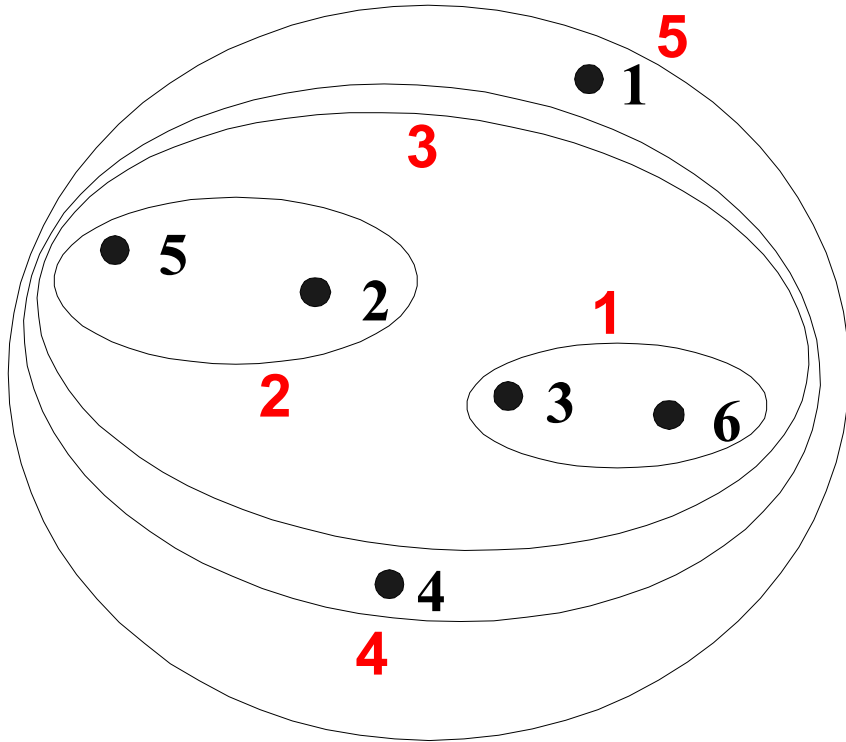
Cluster Similarity: MIN or Single Link

- Similarity of two clusters is based on the two most similar (closest) points in the different clusters
 - Determined by one pair of points, i.e., by one link in the proximity graph.

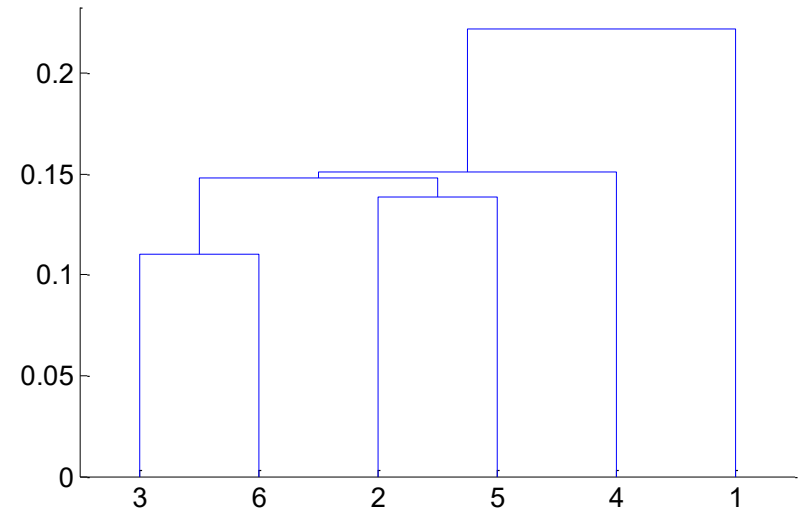
	I1	I2	I3	I4	I5
I1	1.00	0.90	0.10	0.65	0.20
I2	0.90	1.00	0.70	0.60	0.50
I3	0.10	0.70	1.00	0.40	0.30
I4	0.65	0.60	0.40	1.00	0.80
I5	0.20	0.50	0.30	0.80	1.00



Hierarchical Clustering: MIN

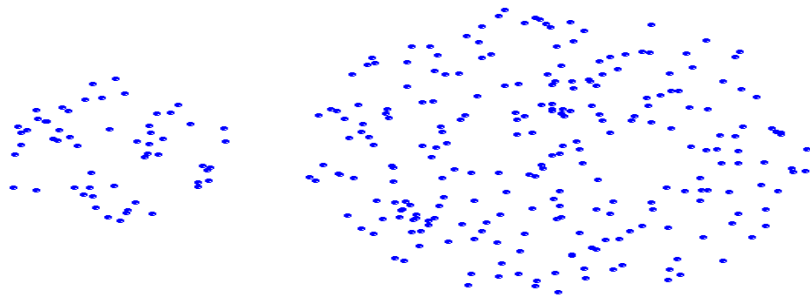


Nested Clusters

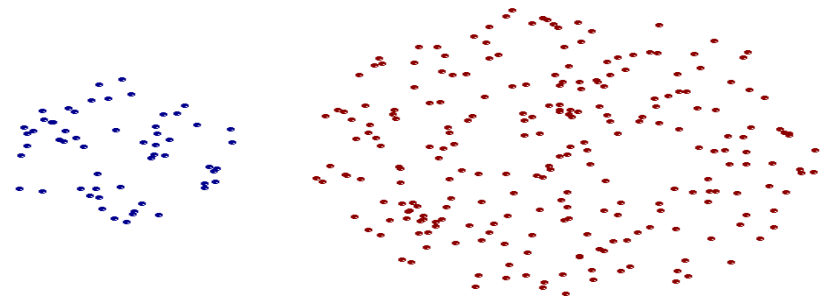


Dendrogram

Strength of MIN



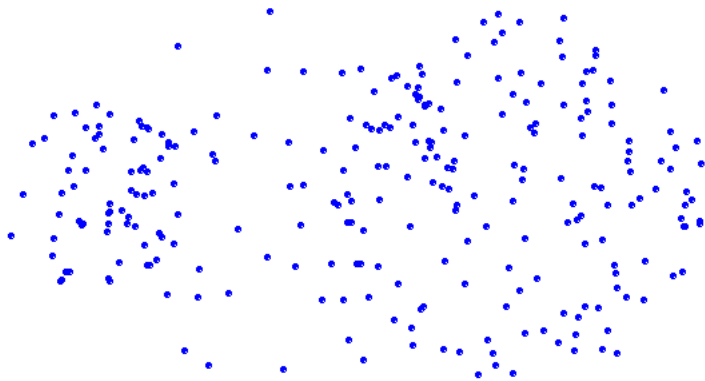
Original Points



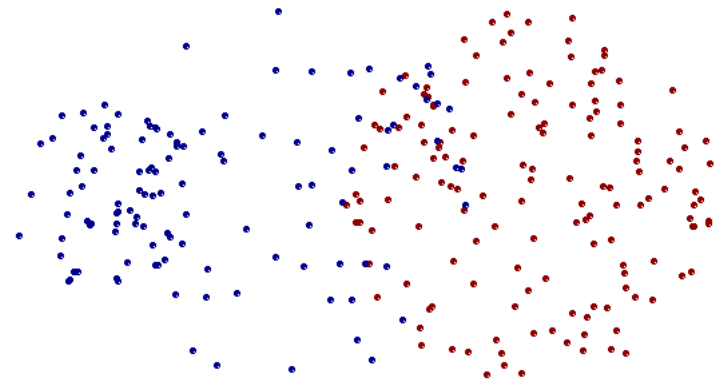
Two Clusters

- **Can handle non-elliptical shapes**

Limitations of MIN



Original Points



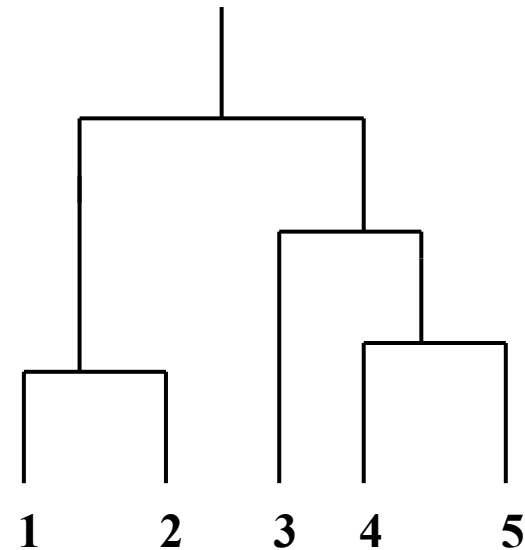
Two Clusters

- **Sensitive to noise and outliers**

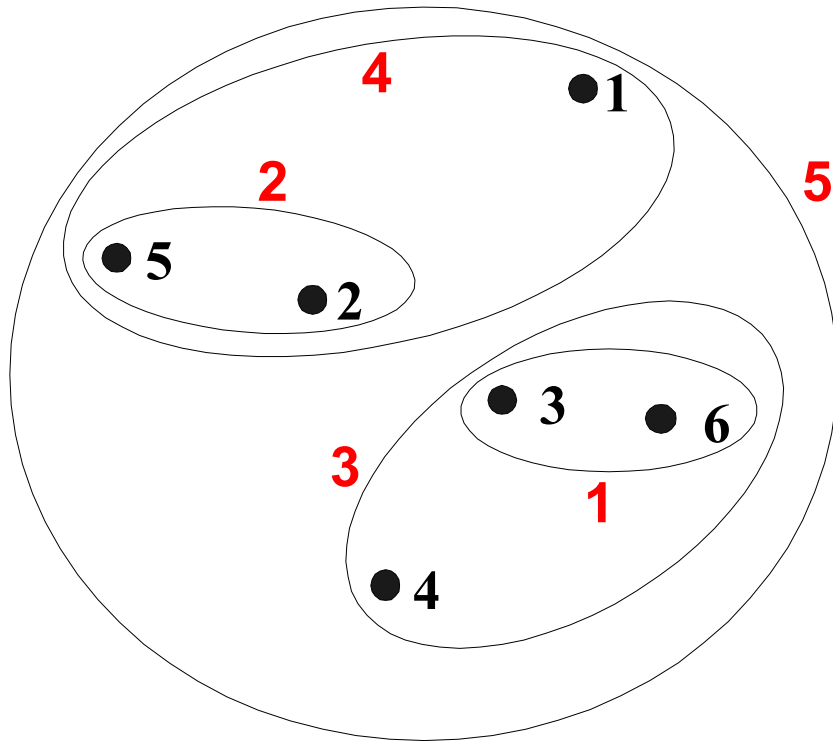
Cluster Similarity: MAX or Complete Linkage

- Similarity of two clusters is based on the two least similar (most distant) points in the different clusters
 - Determined by all pairs of points in the two clusters

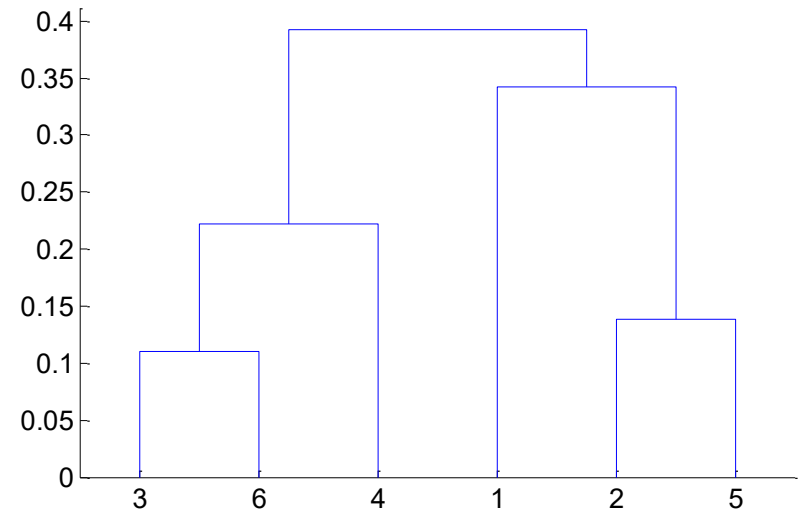
	I1	I2	I3	I4	I5
I1	1.00	0.90	0.10	0.65	0.20
I2	0.90	1.00	0.70	0.60	0.50
I3	0.10	0.70	1.00	0.40	0.30
I4	0.65	0.60	0.40	1.00	0.80
I5	0.20	0.50	0.30	0.80	1.00



Hierarchical Clustering: MAX

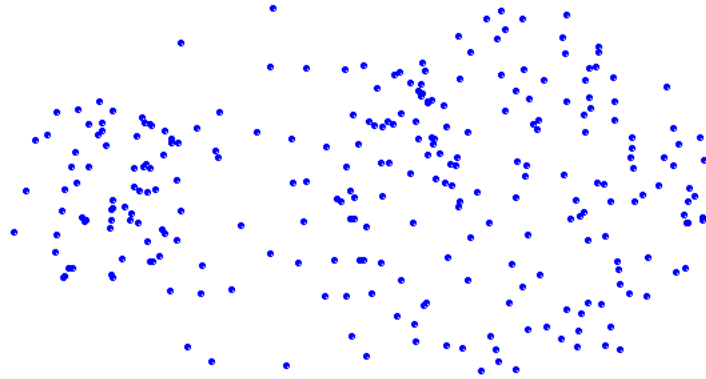


Nested Clusters

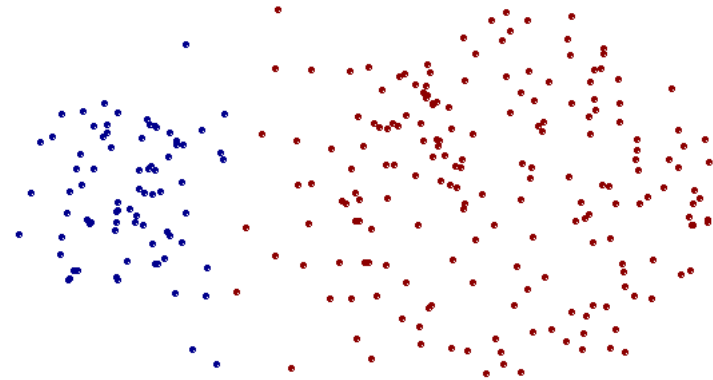


Dendrogram

Strength of MAX



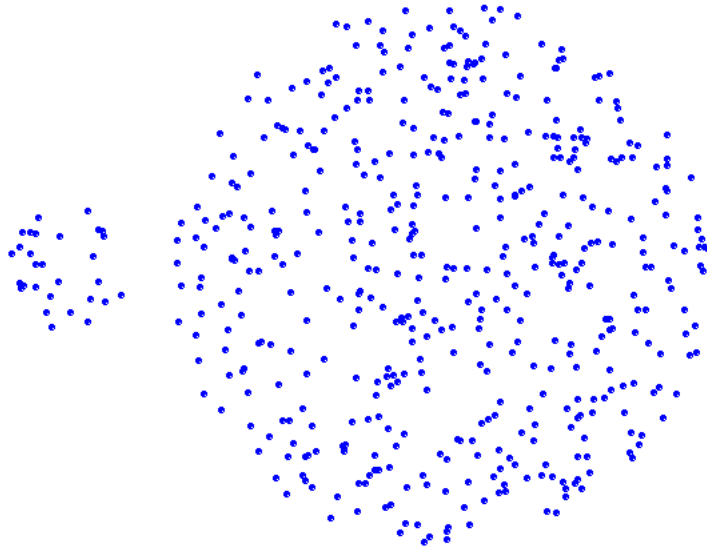
Original Points



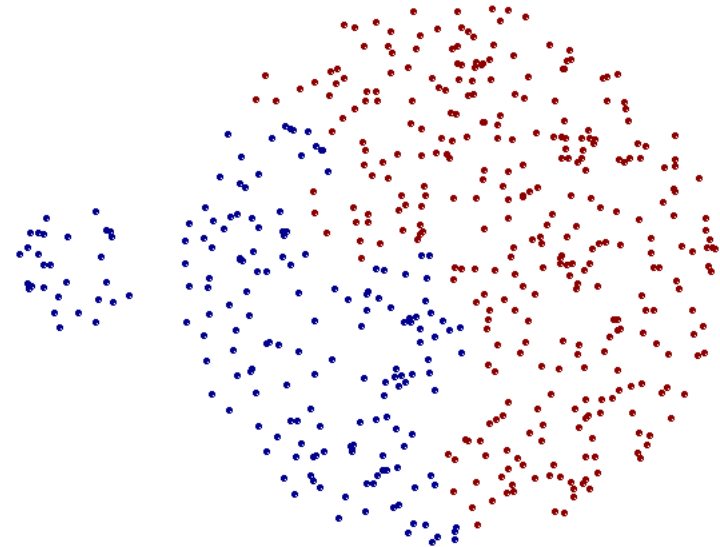
Two Clusters

- **Less susceptible to noise and outliers**

Limitations of MAX



Original Points



Two Clusters

- **Tends to break large clusters**
- **Biased towards globular clusters**

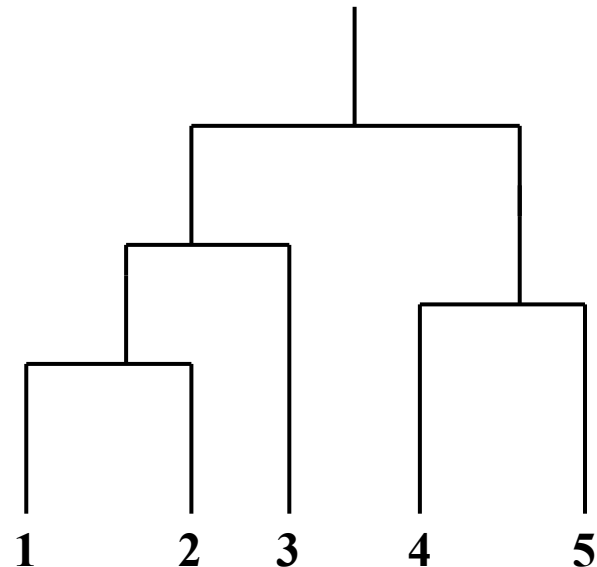
Cluster Similarity: Group Average

- Proximity of two clusters is the average of pairwise proximity between points in the two clusters.

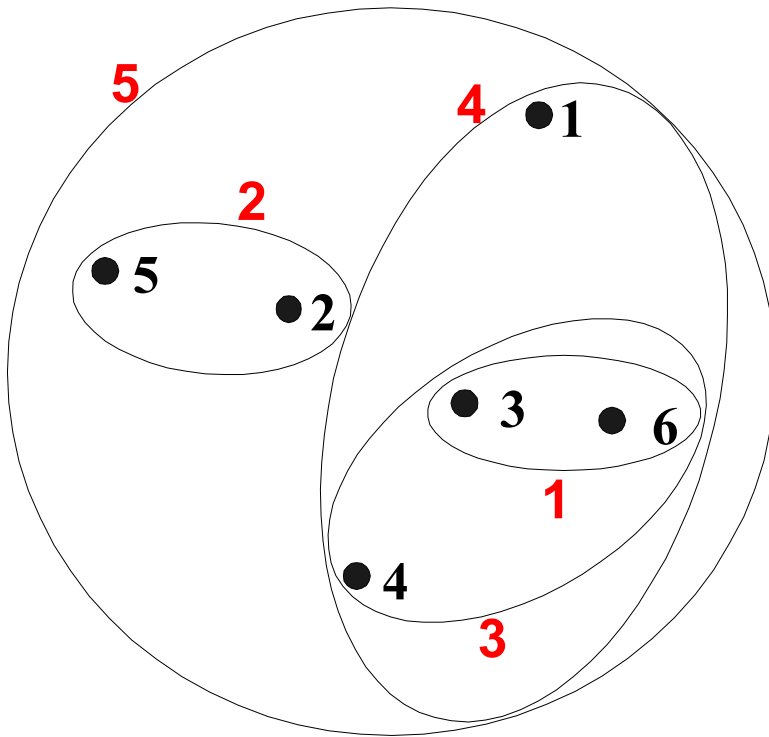
$$\text{proximity}(\text{Cluster}_i, \text{Cluster}_j) = \frac{\sum_{\substack{p_i \in \text{Cluster}_i \\ p_j \in \text{Cluster}_j}} \text{proximity}(p_i, p_j)}{|\text{Cluster}_i| * |\text{Cluster}_j|}$$

- Need to use average connectivity for scalability since total proximity favors large clusters

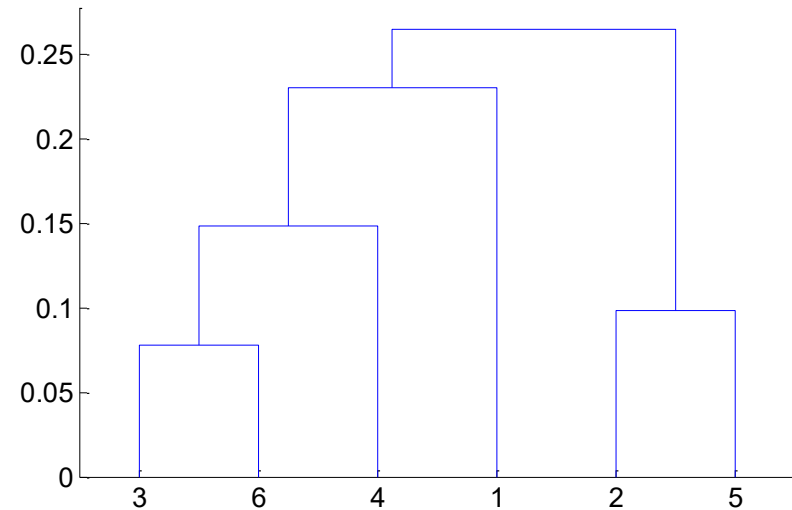
	I1	I2	I3	I4	I5
I1	1.00	0.90	0.10	0.65	0.20
I2	0.90	1.00	0.70	0.60	0.50
I3	0.10	0.70	1.00	0.40	0.30
I4	0.65	0.60	0.40	1.00	0.80
I5	0.20	0.50	0.30	0.80	1.00



Hierarchical Clustering: Group Average



Nested Clusters



Dendrogram

Hierarchical Clustering: Group Average

- Compromise between Single and Complete Link

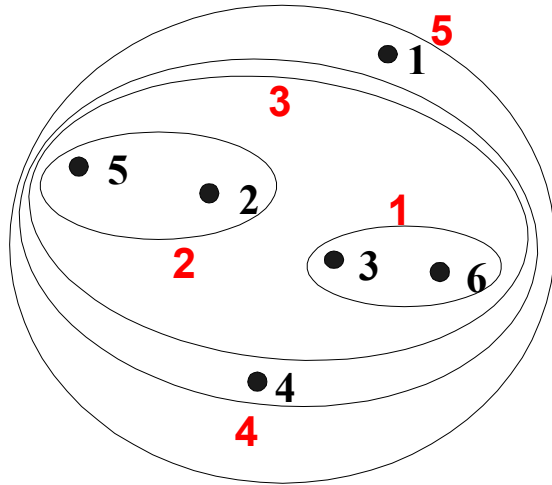
- Strengths
 - Less susceptible to noise and outliers

- Limitations
 - Biased towards globular clusters

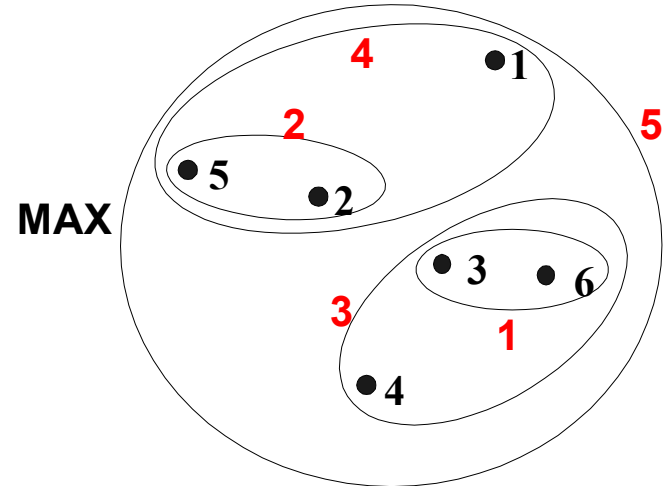
Cluster Similarity: Ward's Method

- Similarity of two clusters is based on the increase in squared error when two clusters are merged
 - Similar to group average if distance between points is distance squared
- Less susceptible to noise and outliers
- Biased towards globular clusters
- Hierarchical analogue of K-means
 - Can be used to initialize K-means

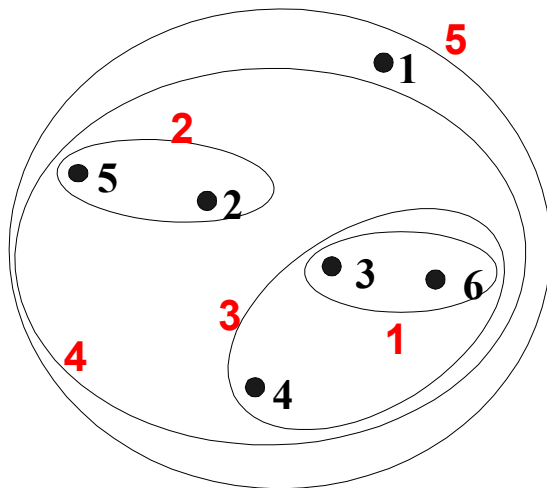
Hierarchical Clustering: Comparison



MIN

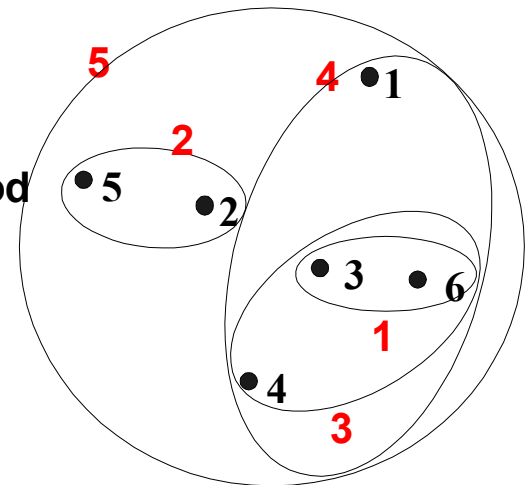


MAX



Group Average

Ward's Method



Hierarchical Clustering: Time and Space requirements

- $O(N^2)$ space since it uses the proximity matrix.
 - N is the number of points.
- $O(N^3)$ time in many cases
 - There are N steps and at each step the size, N^2 , proximity matrix must be updated and searched
 - Complexity can be reduced to $O(N^2 \log(N))$ time for some approaches

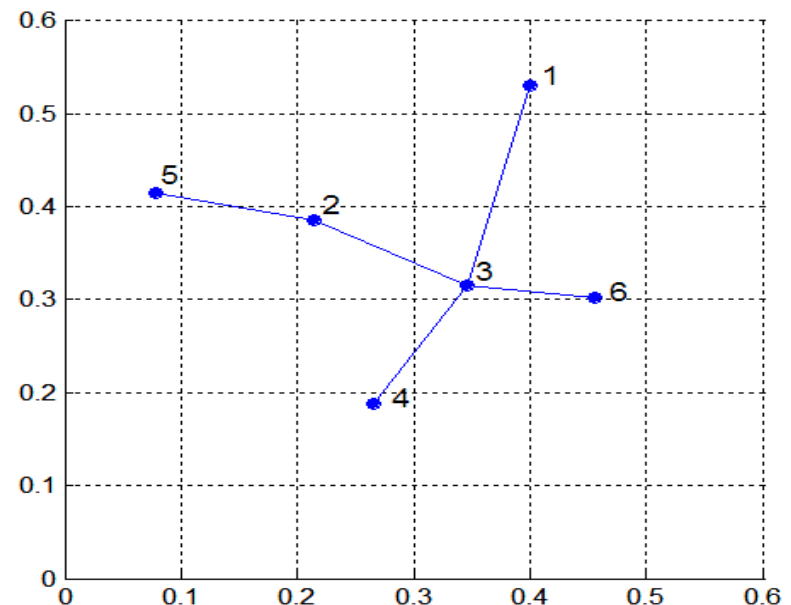
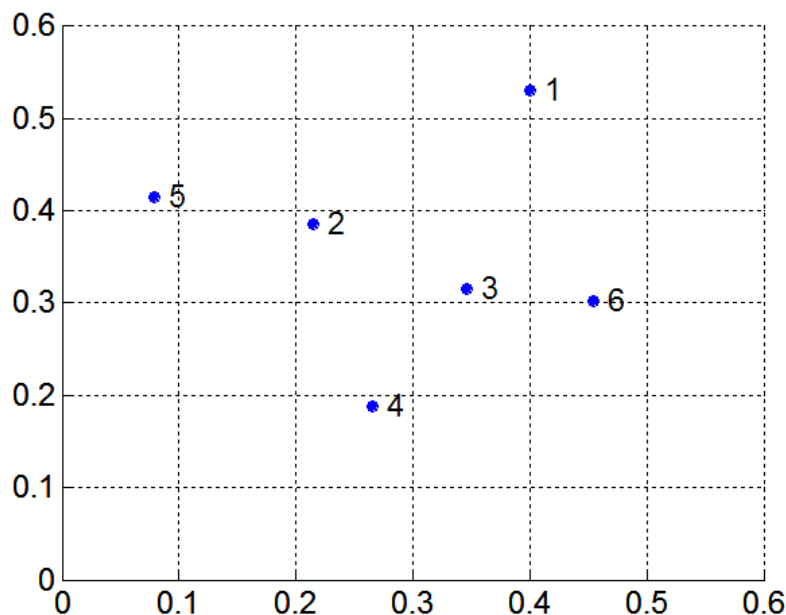
Hierarchical Clustering: Problems and Limitations

- Once a decision is made to combine two clusters, it cannot be undone
- No objective function is directly minimized
- Different schemes have problems with one or more of the following:
 - Sensitivity to noise and outliers
 - Difficulty handling different sized clusters and convex shapes
 - Breaking large clusters

MST: Divisive Hierarchical Clustering

□ Build MST (Minimum Spanning Tree)

- Start with a tree that consists of any point
- In successive steps, look for the closest pair of points (p, q) such that one point (p) is in the current tree but the other (q) is not
- Add q to the tree and put an edge between p and q



MST: Divisive Hierarchical Clustering

- Use MST for constructing hierarchy of clusters

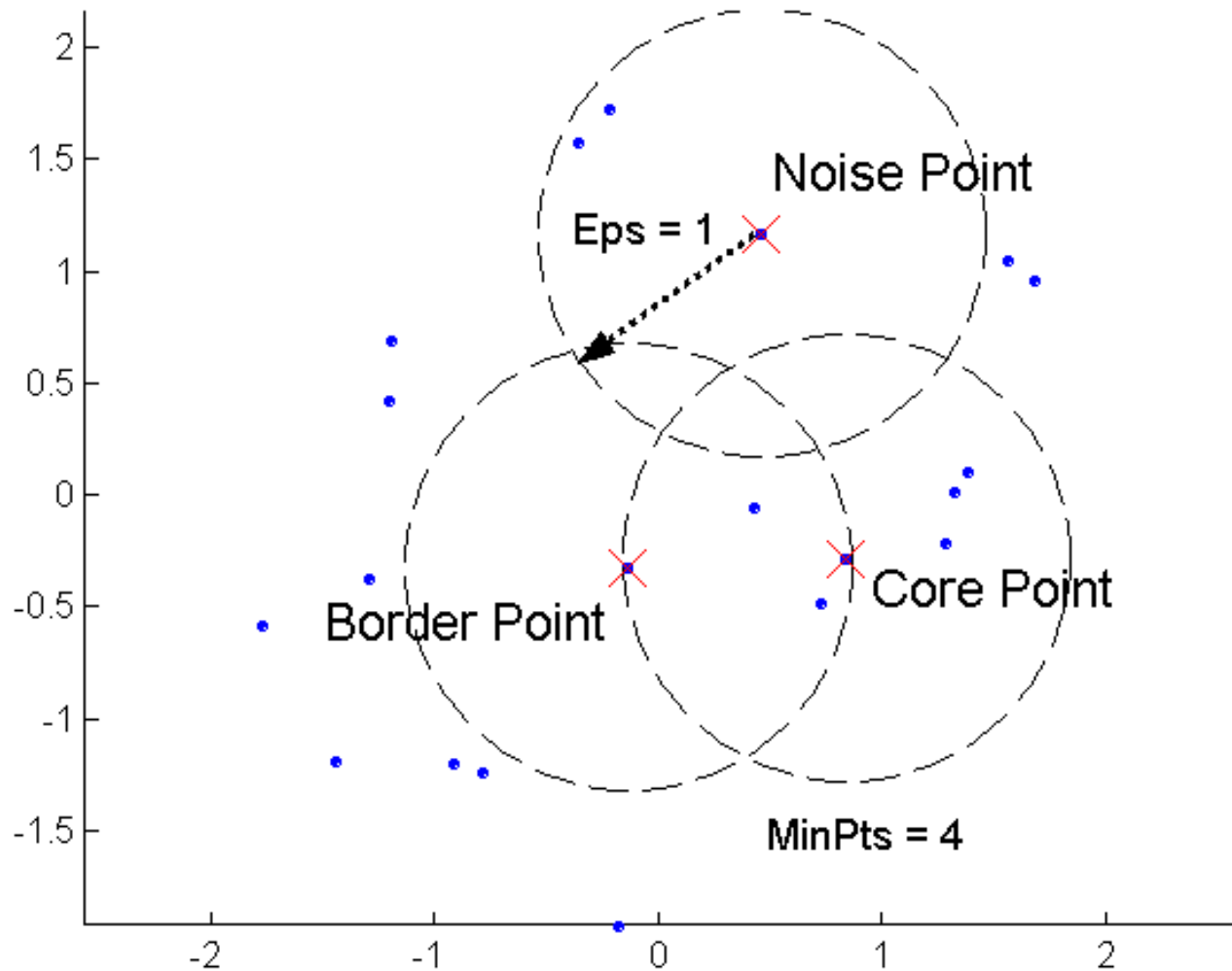
Algorithm 7.5 MST Divisive Hierarchical Clustering Algorithm

- 1: Compute a minimum spanning tree for the proximity graph.
 - 2: **repeat**
 - 3: Create a new cluster by breaking the link corresponding to the largest distance (smallest similarity).
 - 4: **until** Only singleton clusters remain
-

DBSCAN

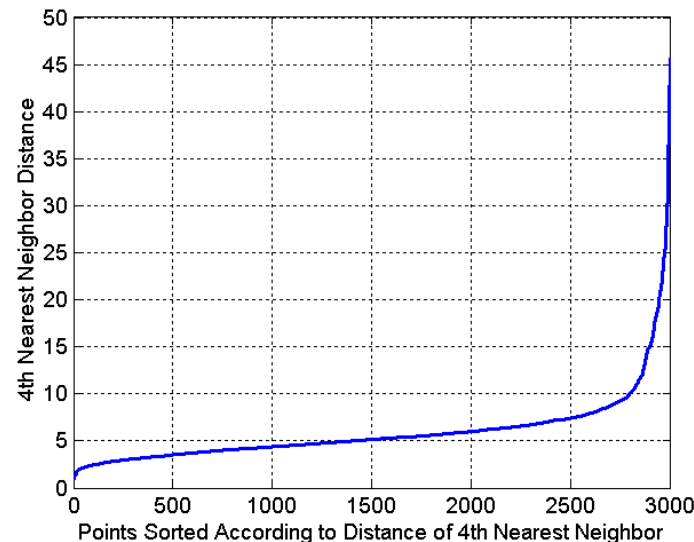
- DBSCAN is a density-based algorithm.
 - Density = number of points within a specified radius (Eps)
 - A point is a **core point** if it has more than a specified number of points (MinPts) within Eps
 - ◆ These are points that are at the interior of a cluster
 - A **border point** has fewer than MinPts within Eps, but is in the neighborhood of a core point
 - A **noise point** is any point that is not a core point or a border point.

DBSCAN: Core, Border, and Noise Points



DBSCAN: Determining EPS and MinPts

- Idea is that for points in a cluster, their k^{th} nearest neighbors are at roughly the same distance
- Noise points have the k^{th} nearest neighbor at farther distance
- So, plot sorted distance of every point to its k^{th} nearest neighbor



DBSCAN Algorithm

- Eliminate noise points
- Perform clustering on the remaining points

$current_cluster_label \leftarrow 1$

for all core points **do**

if the core point has no cluster label **then**

$current_cluster_label \leftarrow current_cluster_label + 1$

 Label the current core point with cluster label $current_cluster_label$

end if

for all points in the Eps -neighborhood, except i^{th} the point itself **do**

if the point does not have a cluster label **then**

 Label the point with cluster label $current_cluster_label$

end if

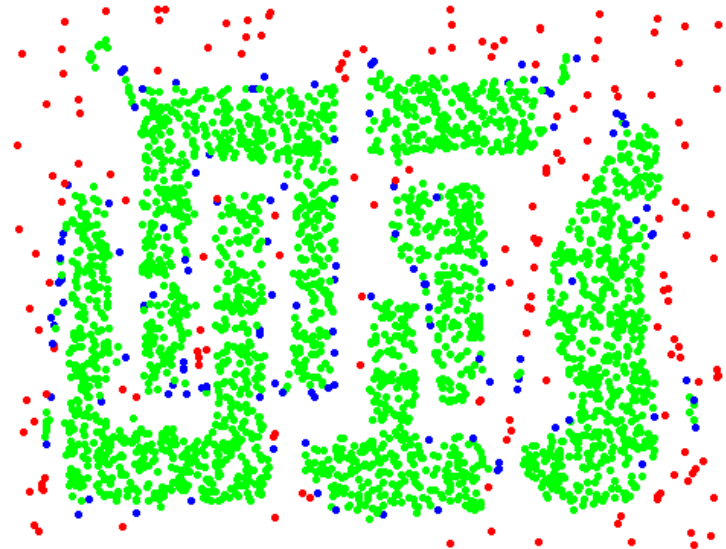
end for

end for

DBSCAN: Core, Border and Noise Points



Original Points



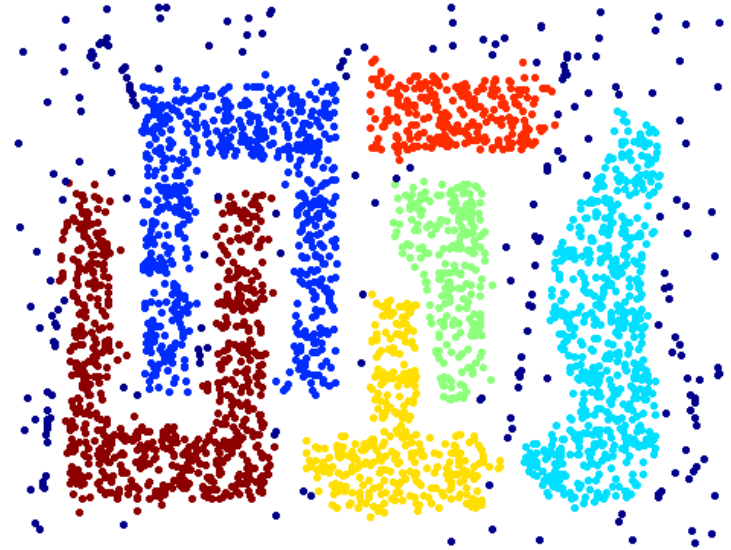
Point types: **core**,
border and **noise**

Eps = 10, MinPts = 4

When DBSCAN Works Well



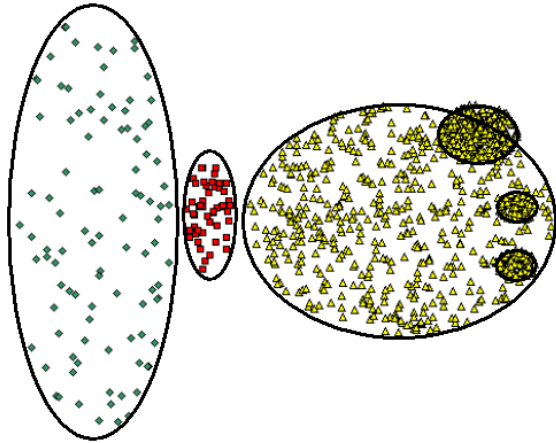
Original Points



Clusters

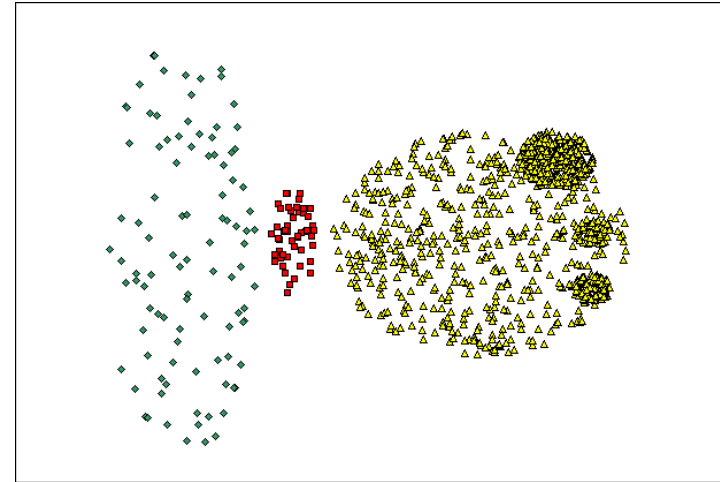
- Resistant to Noise
- Can handle clusters of different shapes and sizes

When DBSCAN Does NOT Work Well

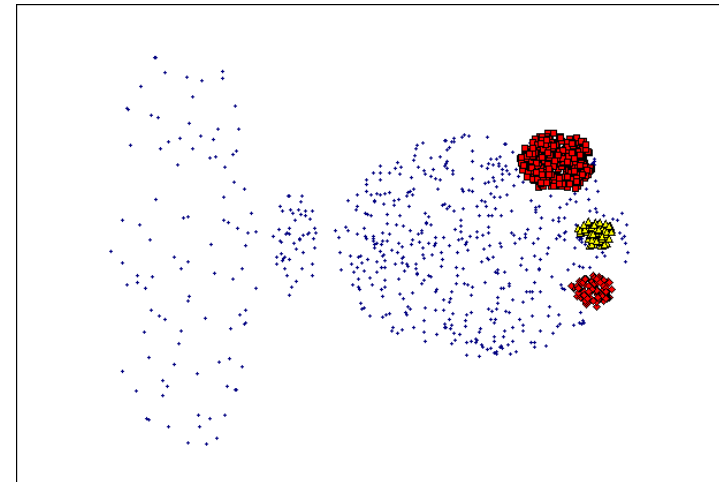


Original Points

- **Varying densities**
- **High-dimensional data**



(MinPts=4, Eps=9.75).



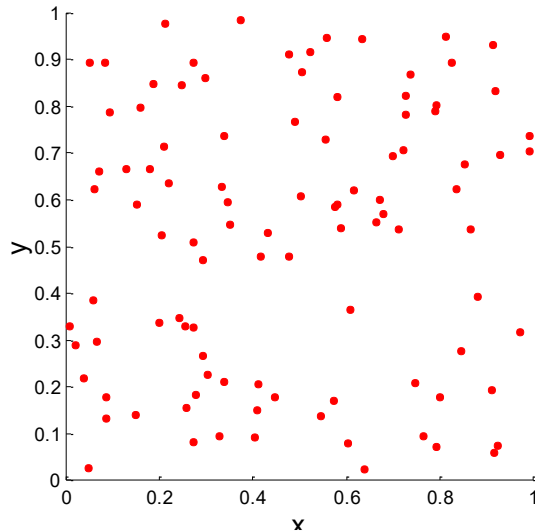
(MinPts=4, Eps=9.92)

Cluster Validity

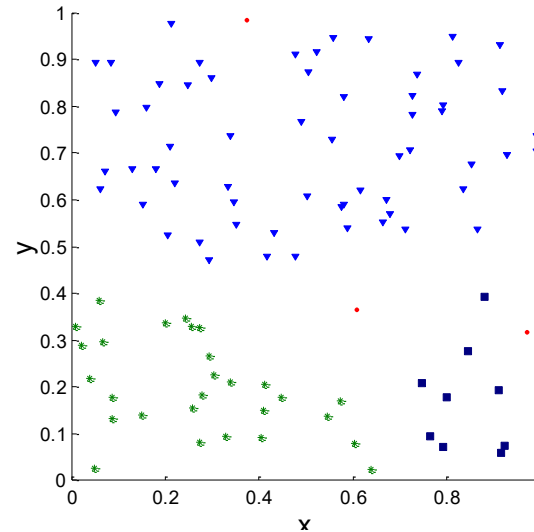
- For supervised classification we have a variety of measures to evaluate how good our model is
 - Accuracy, precision, recall
- For cluster analysis, the analogous question is how to evaluate the “goodness” of the resulting clusters?
- But “clusters are in the eye of the beholder”!
- Then why do we want to evaluate them?
 - To avoid finding patterns in noise
 - To compare clustering algorithms
 - To compare two sets of clusters
 - To compare two clusters

Clusters found in Random Data

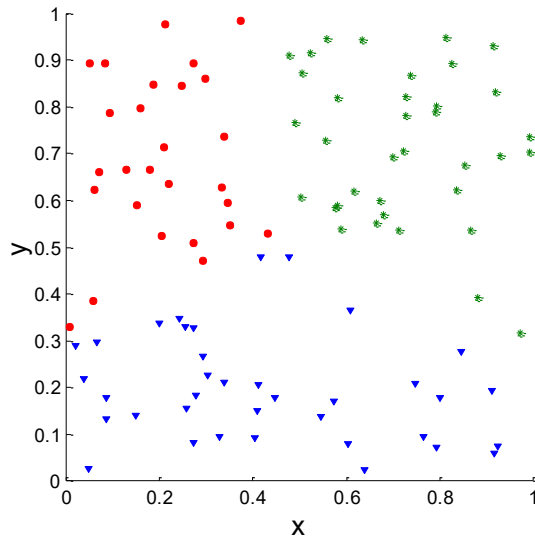
Random Points



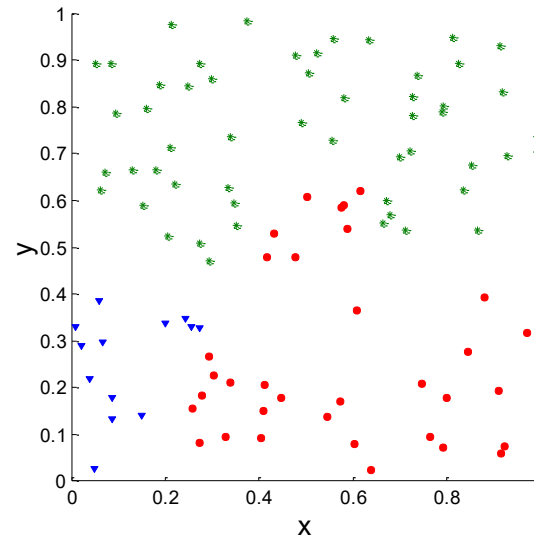
DBSCAN



K-means



Complete Link



Different Aspects of Cluster Validation

1. Determining the **clustering tendency** of a set of data, i.e., distinguishing whether non-random structure actually exists in the data.
2. Comparing the results of a cluster analysis to externally known results, e.g., to externally given class labels.
3. Evaluating how well the results of a cluster analysis fit the data *without* reference to external information.
 - Use only the data
4. Comparing the results of two different sets of cluster analyses to determine which is better.
5. Determining the 'correct' number of clusters.

For 2, 3, and 4, we can further distinguish whether we want to evaluate the entire clustering or just individual clusters.

Measures of Cluster Validity

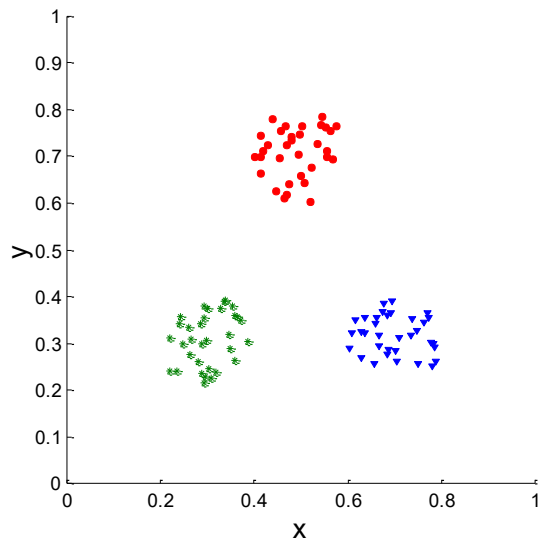
- Numerical measures that are applied to judge various aspects of cluster validity, are classified into the following three types.
 - **External Index:** Used to measure the extent to which cluster labels match externally supplied class labels.
 - ◆ Entropy
 - **Internal Index:** Used to measure the goodness of a clustering structure *without* respect to external information.
 - ◆ Sum of Squared Error (SSE)
 - **Relative Index:** Used to compare two different clusterings or clusters.
 - ◆ Often an external or internal index is used for this function, e.g., SSE or entropy
- Sometimes these are referred to as **criteria** instead of **indices**
 - However, sometimes criterion is the general strategy and index is the numerical measure that implements the criterion.

Measuring Cluster Validity Via Correlation

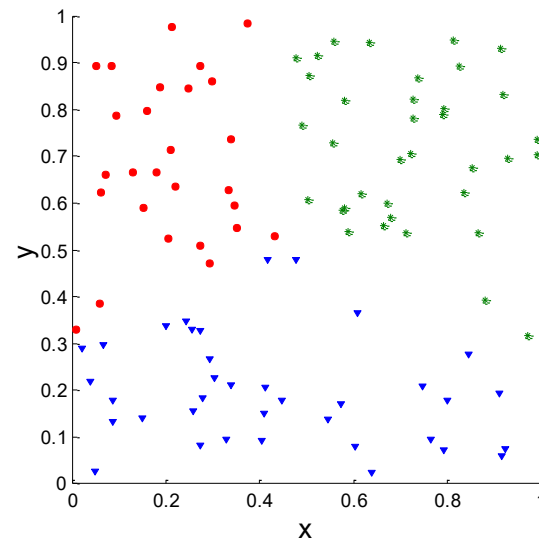
- Two matrices
 - Proximity Matrix
 - “Incidence” Matrix
 - ◆ One row and one column for each data point
 - ◆ An entry is 1 if the associated pair of points belong to the same cluster
 - ◆ An entry is 0 if the associated pair of points belongs to different clusters
- Compute the correlation between the two matrices
 - Since the matrices are symmetric, only the correlation between $n(n-1) / 2$ entries needs to be calculated.
- High correlation indicates that points that belong to the same cluster are close to each other.
- Not a good measure for some density or contiguity based clusters.

Measuring Cluster Validity Via Correlation

- Correlation of incidence and proximity matrices for the K-means clusterings of the following two data sets.



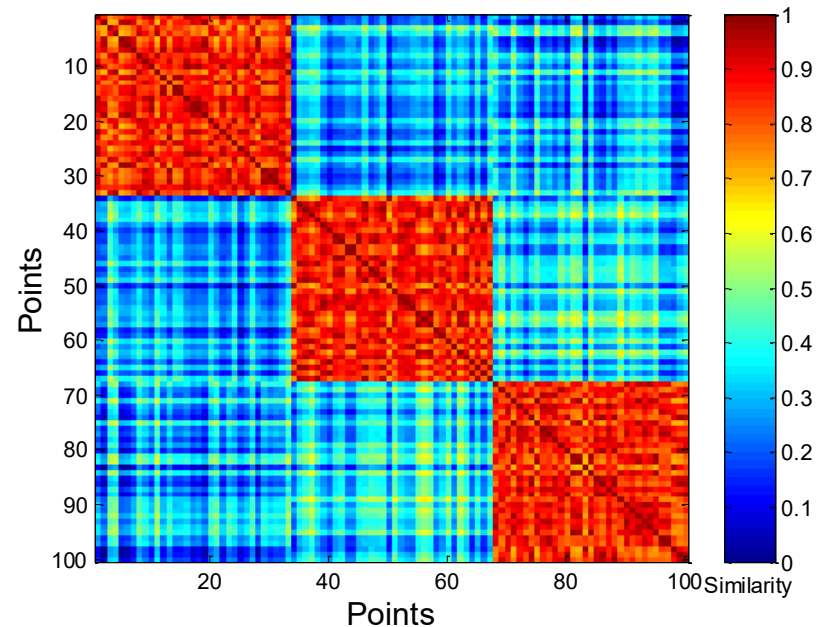
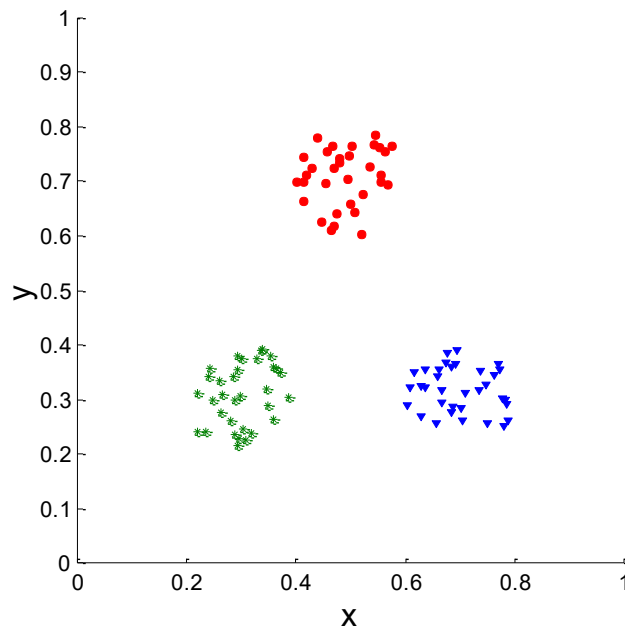
Corr = -0.9235



Corr = -0.5810

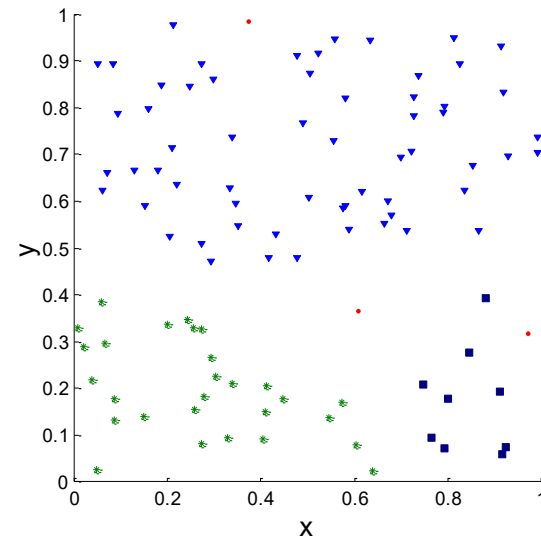
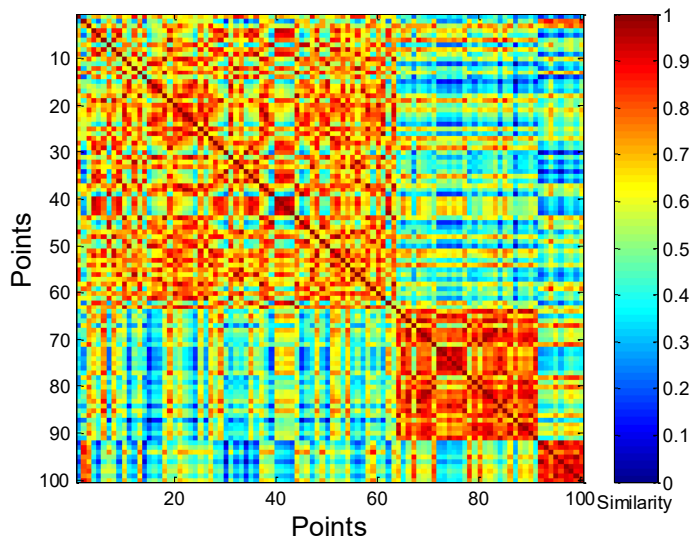
Using Similarity Matrix for Cluster Validation

- Order the similarity matrix with respect to cluster labels and inspect visually.



Using Similarity Matrix for Cluster Validation

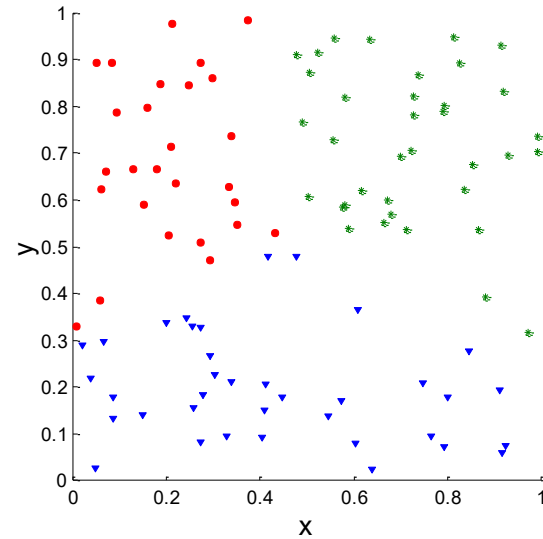
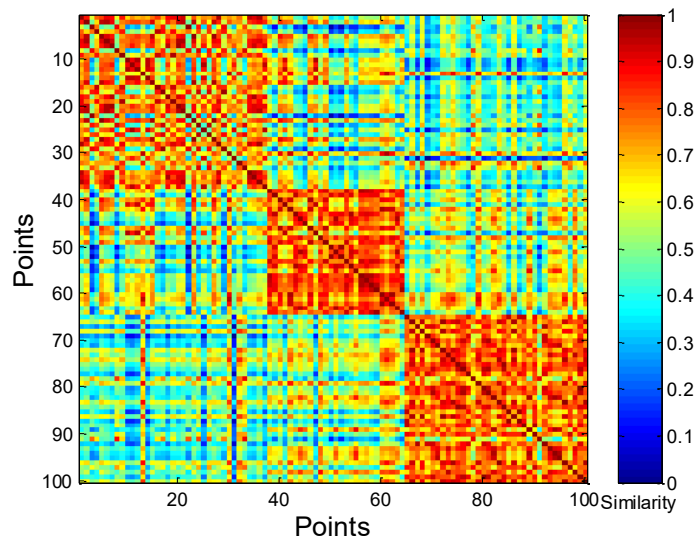
- Clusters in random data are not so crisp



DBSCAN

Using Similarity Matrix for Cluster Validation

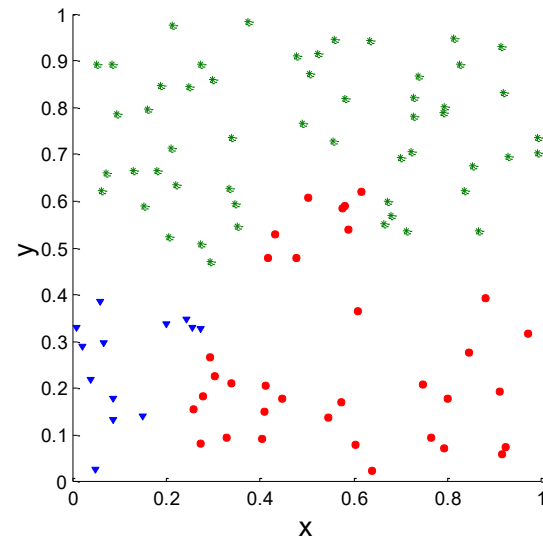
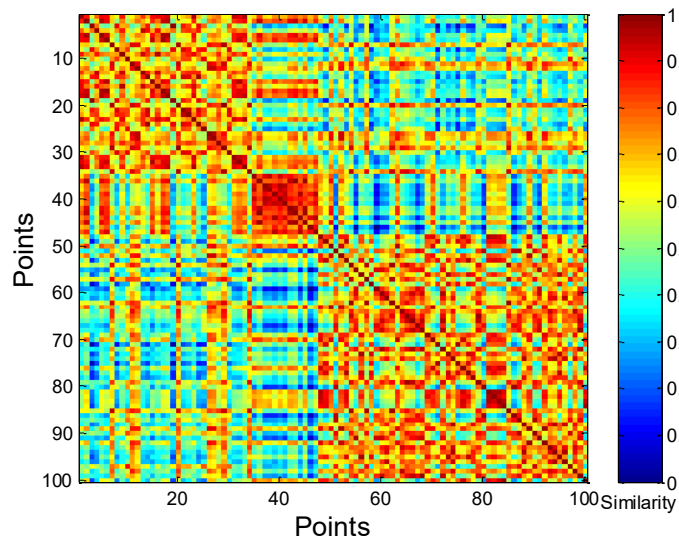
- Clusters in random data are not so crisp



K-means

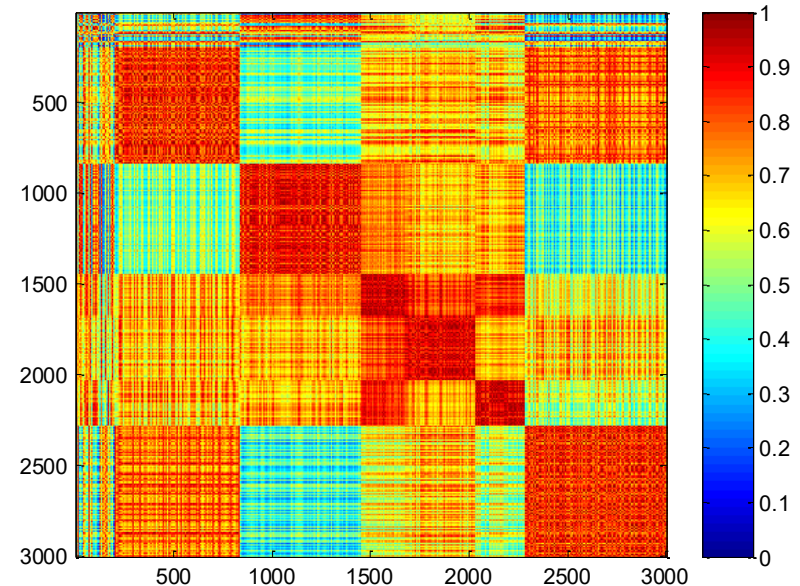
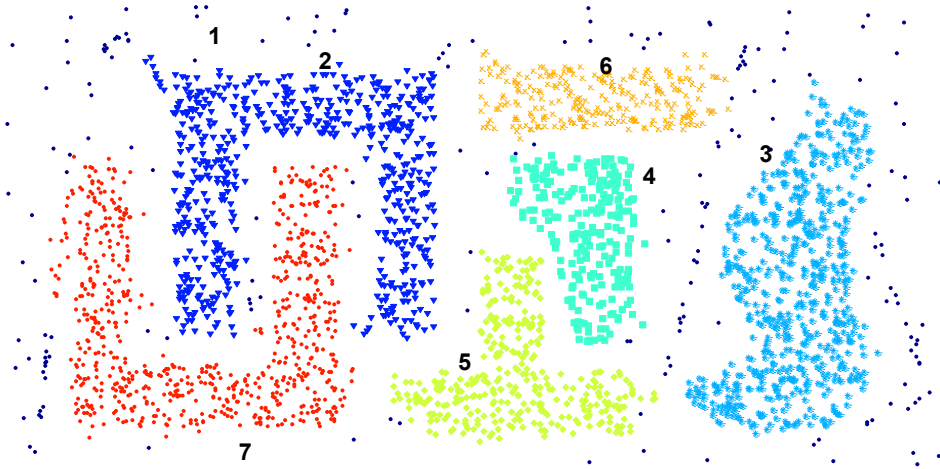
Using Similarity Matrix for Cluster Validation

- Clusters in random data are not so crisp



Complete Link

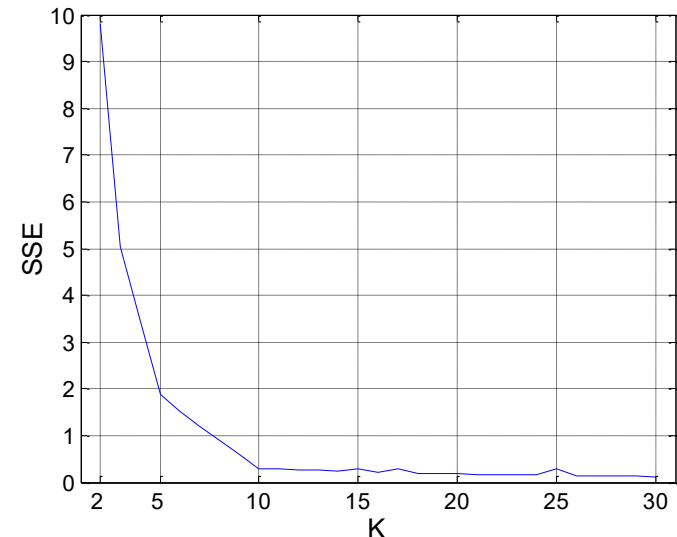
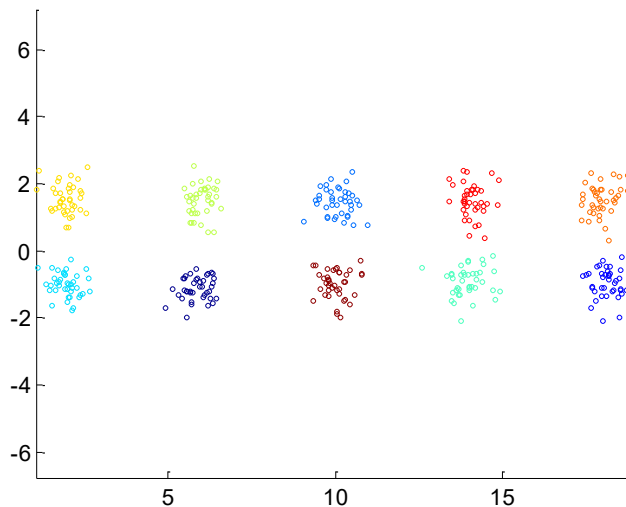
Using Similarity Matrix for Cluster Validation



DBSCAN

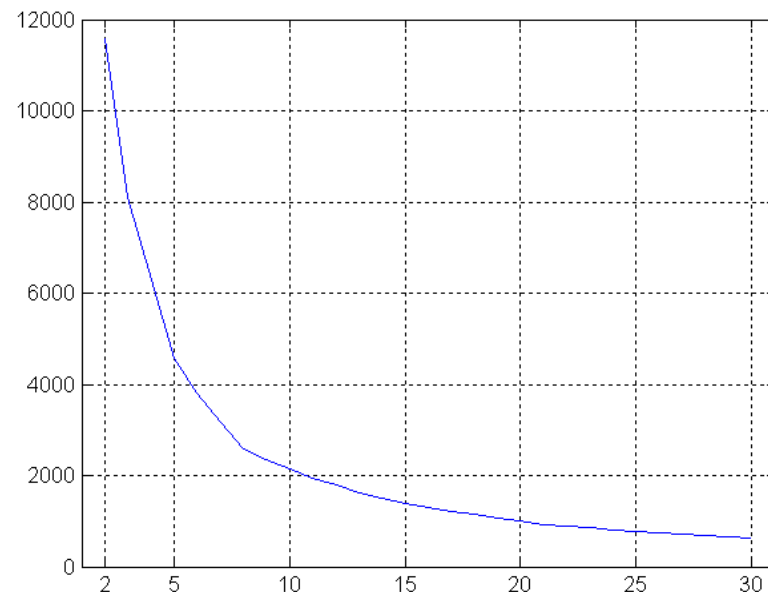
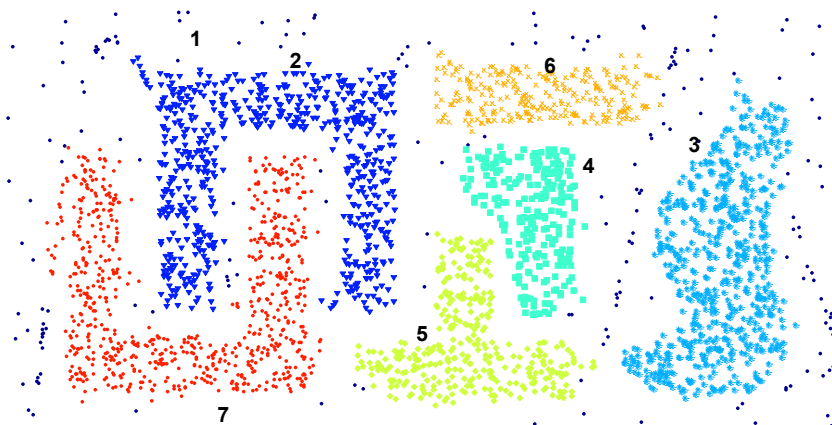
Internal Measures: SSE

- ❑ Clusters in more complicated figures aren't well separated
- ❑ Internal Index: Used to measure the goodness of a clustering structure without respect to external information
 - SSE
- ❑ SSE is good for comparing two clusterings or two clusters (average SSE).
- ❑ Can also be used to estimate the number of clusters



Internal Measures: SSE

- SSE curve for a more complicated data set



SSE of clusters found using K-means

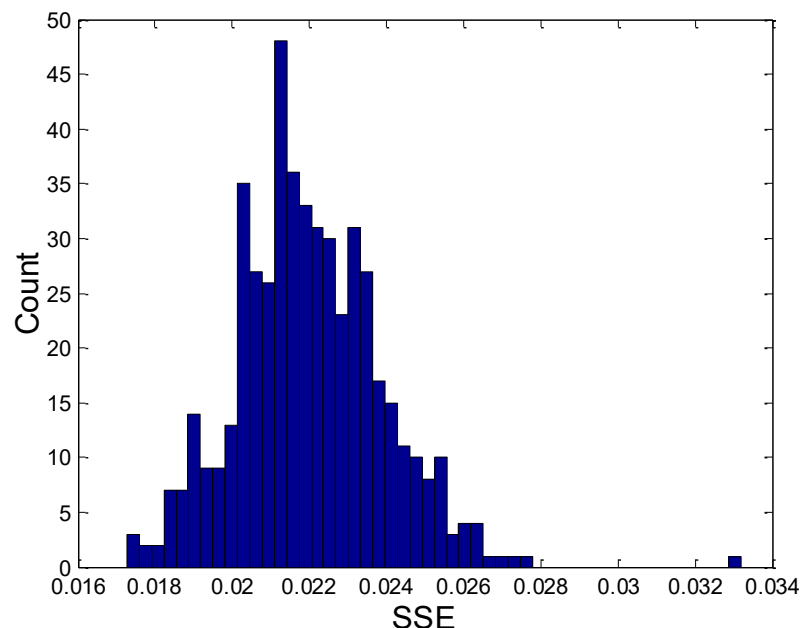
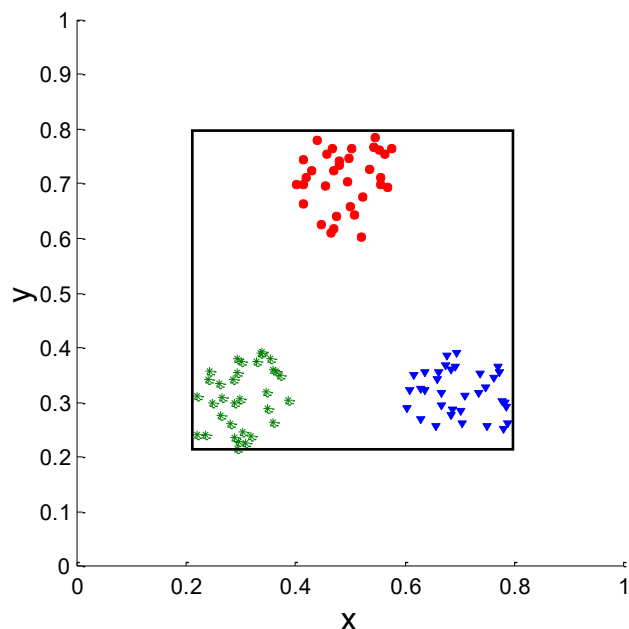
Framework for Cluster Validity

- Need a framework to interpret any measure.
 - For example, if our measure of evaluation has the value, 10, is that good, fair, or poor?
- Statistics provide a framework for cluster validity
 - The more “atypical” a clustering result is, the more likely it represents valid structure in the data
 - Can compare the values of an index that result from random data or clusterings to those of a clustering result.
 - ◆ If the value of the index is unlikely, then the cluster results are valid
 - These approaches are more complicated and harder to understand.
- For comparing the results of two different sets of cluster analyses, a framework is less necessary.
 - However, there is the question of whether the difference between two index values is significant

Statistical Framework for SSE

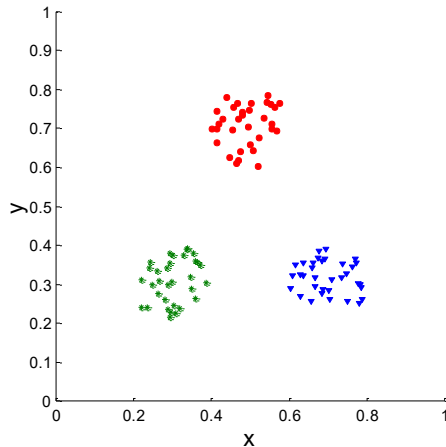
□ Example

- Compare SSE of 0.005 against three clusters in random data
- Histogram shows SSE of three clusters in 500 sets of random data points of size 100 distributed over the range 0.2 – 0.8 for x and y values

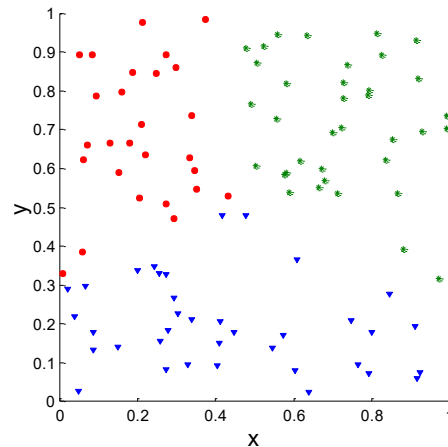


Statistical Framework for Correlation

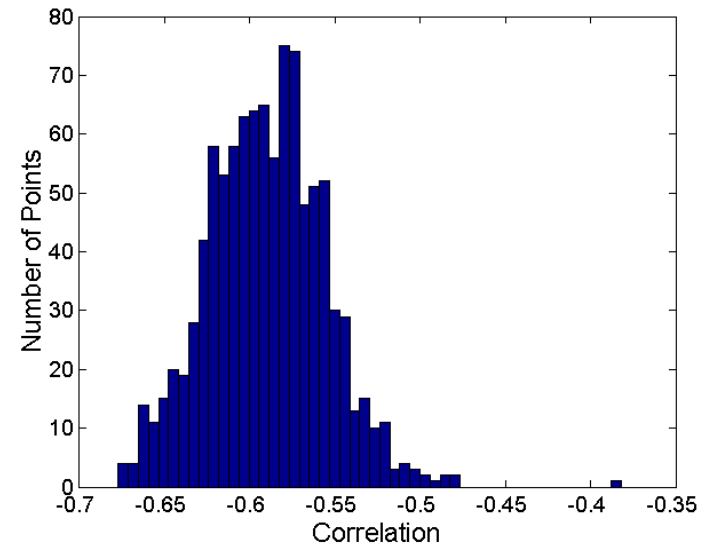
- Correlation of incidence and proximity matrices for the K-means clusterings of the following two data sets.



Corr = -0.9235



Corr = -0.5810



Internal Measures: Cohesion and Separation

- **Cluster Cohesion:** Measures how closely related are objects in a cluster
 - Example: SSE
- **Cluster Separation:** Measure how distinct or well-separated a cluster is from other clusters
- Example: Squared Error
 - Cohesion is measured by the within cluster sum of squares (SSE)
 - Separation is measured by the between cluster sum of squares

$$WSS = \sum_i \sum_{x \in C_i} (x - m_i)^2$$

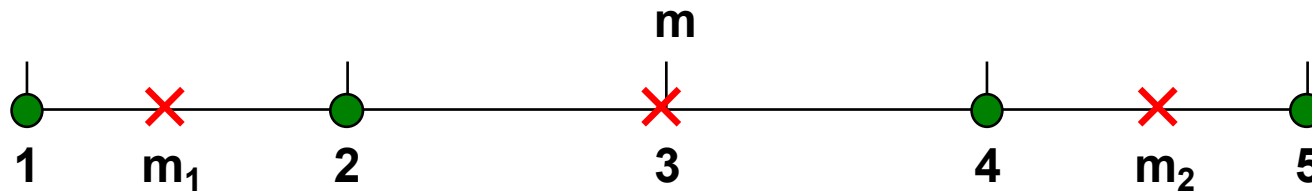
$$BSS = \sum_i |C_i| (m - m_i)^2$$

- Where $|C_i|$ is the size of cluster i

Internal Measures: Cohesion and Separation

□ Example: SSE

— BSS + WSS = constant



K=1 cluster:

$$WSS = (1 - 3)^2 + (2 - 3)^2 + (4 - 3)^2 + (5 - 3)^2 = 10$$

$$BSS = 4 \times (3 - 3)^2 = 0$$

$$Total = 10 + 0 = 10$$

K=2 clusters:

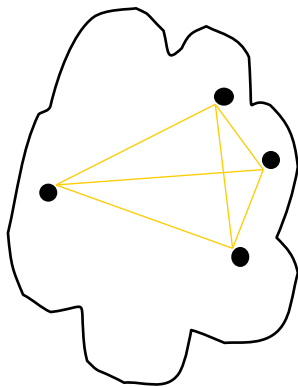
$$WSS = (1 - 1.5)^2 + (2 - 1.5)^2 + (4 - 4.5)^2 + (5 - 4.5)^2 = 1$$

$$BSS = 2 \times (3 - 1.5)^2 + 2 \times (4.5 - 3)^2 = 9$$

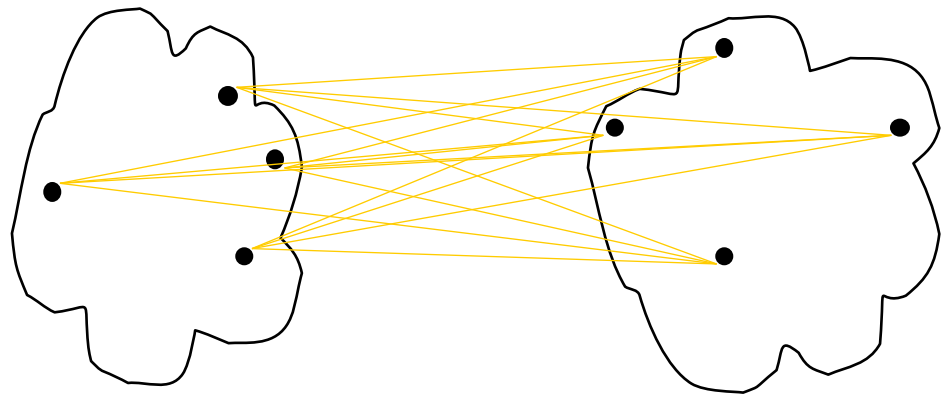
$$Total = 1 + 9 = 10$$

Internal Measures: Cohesion and Separation

- A proximity graph based approach can also be used for cohesion and separation.
 - Cluster cohesion is the sum of the weight of all links within a cluster.
 - Cluster separation is the sum of the weights between nodes in the cluster and nodes outside the cluster.



cohesion



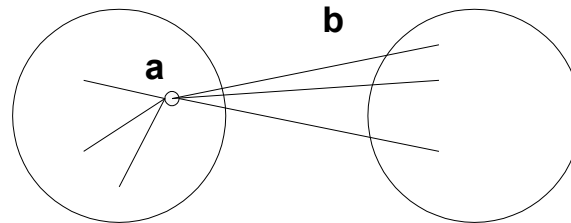
separation

Internal Measures: Silhouette Coefficient

- Silhouette Coefficient combine ideas of both cohesion and separation, but for individual points, as well as clusters and clusterings
- For an individual point, i
 - Calculate a = average distance of i to the points in its cluster
 - Calculate b = min (average distance of i to points in another cluster)
 - The silhouette coefficient for a point is then given by

$$s = 1 - a/b \quad \text{if } a < b, \quad (\text{or } s = b/a - 1 \quad \text{if } a \geq b, \text{ not the usual case})$$

- Typically between 0 and 1.
- The closer to 1 the better.



- Can calculate the Average Silhouette width for a cluster or a clustering

External Measures of Cluster Validity: Entropy and Purity

Table 5.9. K-means Clustering Results for LA Document Data Set

Cluster	Entertainment	Financial	Foreign	Metro	National	Sports	Entropy	Purity
1	3	5	40	506	96	27	1.2270	0.7474
2	4	7	280	29	39	2	1.1472	0.7756
3	1	1	1	7	4	671	0.1813	0.9796
4	10	162	3	119	73	2	1.7487	0.4390
5	331	22	5	70	13	23	1.3976	0.7134
6	5	358	12	212	48	13	1.5523	0.5525
Total	354	555	341	943	273	738	1.1450	0.7203

entropy For each cluster, the class distribution of the data is calculated first, i.e., for cluster j we compute p_{ij} , the ‘probability’ that a member of cluster j belongs to class i as follows: $p_{ij} = m_{ij}/m_j$, where m_j is the number of values in cluster j and m_{ij} is the number of values of class i in cluster j . Then using this class distribution, the entropy of each cluster j is calculated using the standard formula $e_j = \sum_{i=1}^L p_{ij} \log_2 p_{ij}$, where the L is the number of classes. The total entropy for a set of clusters is calculated as the sum of the entropies of each cluster weighted by the size of each cluster, i.e., $e = \sum_{j=1}^K \frac{m_j}{m} e_j$, where m_j is the size of cluster j , K is the number of clusters, and m is the total number of data points.

purity Using the terminology derived for entropy, the purity of cluster j , is given by $purity_j = \max_i p_{ij}$ and the overall purity of a clustering by $purity = \sum_{j=1}^K \frac{m_j}{m} purity_j$.

Final Comment on Cluster Validity

“The validation of clustering structures is the most difficult and frustrating part of cluster analysis.

Without a strong effort in this direction, cluster analysis will remain a black art accessible only to those true believers who have experience and great courage.”

Algorithms for Clustering Data, Jain and Dubes